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Anaphylaxis: Emergency treatment

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INTRODUCTION

Anaphylaxis is a serious, systemic, mast cell-mediated event that can be fatal if not promptly recognized and treated. The goal of therapy is early recognition and treatment with epinephrine to prevent progression to life-threatening respiratory and/or cardiovascular symptoms and signs, including asphyxiation and shock.

Anaphylaxis is variable and unpredictable. It may be mild and resolve spontaneously due to endogenous production of compensatory mediators. On the other end of the spectrum, it may be severe and progress within minutes to respiratory or cardiovascular compromise and death [1]. At the onset of an anaphylactic episode, it is not possible to predict how severe it will become, how rapidly it will progress, and whether it will resolve promptly and completely, because the factors that determine the course of anaphylaxis in an individual patient are not fully understood. Because of these variables, it is important to administer intramuscular (IM) epinephrine as soon as anaphylaxis is recognized or if impending anaphylaxis is suspected, even if patients do not meet diagnostic criteria.

This topic will discuss the treatment of anaphylaxis. The clinical manifestations and diagnosis of anaphylaxis, pathophysiology, and unique features of anaphylaxis in specific patient groups are reviewed separately:

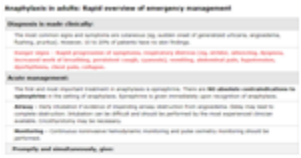
- (See "Anaphylaxis: Acute diagnosis".)
- (See "Biphasic and protracted anaphylaxis".)
- (See "Pathophysiology of anaphylaxis".)
- (See "Fatal anaphylaxis".)

- (See "Anaphylaxis: Confirming the diagnosis and determining the cause(s)".)
- (See "Anaphylaxis in infants".)
- (See "Anaphylaxis during pregnancy, delivery, and lactation".)

Recommendations for dosing and prescribing epinephrine for use by patients and caregivers in the community setting are provided separately. (See "Prescribing epinephrine for anaphylaxis self-treatment", section on 'Dosing'.)

INITIAL ASSESSMENT AND MANAGEMENT

The cornerstone of initial management includes critical components that need to be instituted concomitantly [2-13]. Rapid overview tables that summarize important interventions in the first few minutes of management are provided for adults (table 1) and for infants and children (table 2).



Anaphylaxis in adults: Rapid overview of emergency management

Table 1 - larger image below

Anaphylaxis in infants and children: Rapid overview of emergency management*

Initial stabilization — The following steps are often performed simultaneously, depending on available resources and symptom acuity. Once anaphylaxis is recognized, epinephrine should be administered as soon as possible. (See 'Epinephrine injection' below.)

- **Call for help** – Summon a resuscitation team in a hospital setting or call 911 or an equivalent emergency medical services number with the clear complaint of "anaphylaxis" in a community setting. If the patient is not in an acute care setting at the time epinephrine is administered, the 911 response should be initiated without waiting to see if the reaction worsens. The 911 call can be aborted if the patient recovers, or the patient can be assessed by paramedics and not transported.
- **Assess airway, breathing, and circulation** – Simultaneously with epinephrine injection, attention should focus first on airway, breathing, and circulation, as well as adequacy of mentation. The lips, tongue, and oral pharynx are assessed

Table 2 - larger image below

for angioedema, and the patient is asked to speak their name to assess periglottic or glottic swelling. Stridor is an ominous finding and represents advanced airway compromise with potentially imminent airway obstruction. Prepare to establish an airway if required. (See 'Airway management' below.)

- **Administer oxygen** – Supplemental oxygen, initially using a nonrebreather mask at 15 liters/minute flow rate or commercial high-flow oxygen masks (providing at least 70% and up to 100% oxygen), should be administered until it is known that there is no significant bronchospasm or evolving upper airway obstruction.

- **Prompt recognition** – Prompt assessment and treatment are critical in any patient suspected of having anaphylaxis (table 3 and table 4) as respiratory or cardiac arrest and death can occur within minutes [14-22]. Examine the skin for urticaria or angioedema, which (if present) is helpful in confirming the diagnosis (table 5). Severe anaphylaxis can progress to death within minutes. In a series of 164 fatalities due to anaphylaxis, the median time interval between onset of symptoms and respiratory or cardiac arrest was 5 minutes in iatrogenic anaphylaxis, 15 minutes in stinging insect venom-induced anaphylaxis, and 30 minutes in food-induced anaphylaxis [14]. Anaphylaxis appears to be most responsive to treatment in its early phases, based on the observation that delayed epinephrine injection is associated with increased fatalities [23-27]. (See 'Efficacy' below.)

- **Remove inciting cause (if possible)** – As an example, stop the infusion of a suspect medication. There is no evidence that gastric emptying has any benefit in cases of ingested food allergens.

- **Position the patient** – If the patient is hypotensive, place them in the recumbent position with the lower extremities elevated, unless there is prominent upper airway swelling or bronchospasm prompting the patient to remain upright (and often leaning forward). If the patient is vomiting, placement in the semirecumbent position with lower extremities elevated may be preferable. Place pregnant patients on their left side to minimize compression of the inferior vena cava by the gravid uterus. The recumbent position maximizes perfusion of vital organs and helps prevent severe hypotension, subsequent inadequate cardiac filling, and pulseless cardiac activity [28]. If patients are kept upright or moved to an upright posture in the setting of hypoperfusion, death can occur within seconds (ie, "empty ventricle syndrome") [7].

- **Establish IV access** – For hypotensive patients, two large-bore intravenous (IV) catheters (ideally 14 to 16 gauge for most adults) should be inserted in preparation for rapid administration of fluids and medications. Intraosseous access should be obtained if IV access is not readily obtainable.

- **Continuously monitor** – Continuous electronic monitoring of cardiopulmonary status, including frequent measurements of blood pressure (BP), heart rate, and respiratory rate, as well as monitoring of oxygen saturation by pulse oximetry, is required for the duration of the episode.

Epinephrine injection — At the earliest opportunity, epinephrine should be injected intramuscularly (IM) into the mid-anterolateral aspect of the thigh (table 1 and table 2), followed by additional epinephrine as needed [3-7,23,29,30]. If symptoms are severe, an IV epinephrine infusion should be prepared in case it is needed. (See 'Intramuscular epinephrine injection (preferred)' below.)

Until more data are available on the performance of epinephrine nasal spray in patients with anaphylaxis, we prefer IM administration of epinephrine in most situations. (See 'Epinephrine nasal spray' below.)

Airway management — The initial steps in anaphylaxis management involve a rapid assessment of the patient's airway [8-10,12,13]:

- Tracheal intubation should be performed on an emergency basis if stridor or respiratory distress due to upper airway compromise is present. (See "Approach to the anatomically difficult airway in adults for emergency medicine and critical care" and "The difficult pediatric airway for emergency medicine".)

- Preparations for early intubation should be made if there is any airway involvement or significant edema of the tongue or oropharyngeal tissues, including the uvula, or if voice alteration has occurred, especially if only a small amount of time has elapsed since the exposure. Early presence of upper airway edema represents rapidly developing airway compromise, requiring prompt action. In a patient who does not require prompt tracheal intubation, a flexible nasopharyngolaryngoscopy airway examination ("awake look") is an option to exclude laryngeal edema, the presence of which would prompt definitive airway management. (See "Approach to the anatomically difficult airway in adults for emergency medicine and critical care", section on 'Awake airway examination'.)

- An emergency cricothyroidotomy may be required to secure the airway if severe upper airway edema prevents access to the glottic aperture, even with use of a video laryngoscope. In patients with severe airway edema, a cricothyrotomy may be the primary airway technique if other techniques are felt unlikely to be successful or too risky. Failure to rapidly perform a cricothyrotomy in a patient with severe upper airway edema in a "can't intubate, can't oxygenate" situation may result in anoxic encephalopathy and/or death. Use of an extraglottic airway device (eg, laryngeal mask airway) may be attempted as a rescue maneuver or as a bridge to provide ventilation while making preparations for cricothyrotomy, but these devices are dependent on normal upper airway anatomy so may not be successful. (See "Emergency cricothyrotomy (cricothyroidotomy)".)

Tracheal intubation may be difficult in individuals in whom edema distorts airway anatomical landmarks (algorithm 1). A "triple setup" should be used, which implies that a primary intubating plan is in place followed by a nonsurgical rescue device (eg, an extraglottic device) and lastly a surgical rescue plan, most often an open surgical bougie-assisted cricothyrotomy. Unless unavailable, a video laryngoscope should be used for all intubations with distorted upper airway anatomy. In patients with rapidly worsening airway edema (ie, "forced to act scenario"), a prompt decision to administer rapid sequence intubation medications and create the best possible situation for a **single best attempt** at tracheal intubation, whether by laryngoscopy or surgical airway, often is preferable to considering other approaches. If this single best attempt is not successful, a failed airway is present (algorithm 2). Failed attempts at tracheal intubation can lead to complete airway obstruction and fatality. Therefore, upper airway closure in the setting of anaphylaxis should be managed by the most experienced clinician available. This may require immediate collaboration between an emergency physician and an anesthesiologist, otolaryngologist, or intensivist with training and experience in the management of the difficult airway, but delay should not occur while seeking consultation. If airway assistance is not immediately available, tracheal intubation should ensue. (See "Approach to the anatomically difficult airway in adults for emergency medicine and critical care" and "The difficult pediatric airway for emergency medicine".)

Intravenous fluids — Fluid resuscitation should be initiated immediately in patients who present with hypotension or incomplete response to IM epinephrine. Massive fluid shifts can occur rapidly due to increased vascular permeability, with transfer of up to 35 percent of the intravascular volume into the extravascular space within minutes [3]. Any patient whose hypotension does not respond promptly and completely to IM epinephrine should receive large-volume fluid resuscitation [8-10,12,13]:

- Adults should receive 1 to 2 liters of a crystalloid solution at the most rapid flow rate possible in the first minutes of treatment. Large volumes of fluid (eg, up to 7 liters) may be required.
- Children should receive crystalloid solution in boluses of 20 mL/kg, each over 5 to 10 minutes, and repeated, as needed. Large volumes of fluid (up to 100 mL/kg) may be required [31].

Patients should be monitored carefully and continuously for clinical response and for volume overload. (See "Treatment of severe hypovolemia or hypovolemic shock in adults" and "Hypovolemic shock in children in resource-abundant settings: Initial evaluation and management".)

Considerations during pregnancy — Additional precautions and considerations are important in the management of anaphylaxis during pregnancy. For example, during labor and delivery, positioning of the patient on their left side, providing high-flow supplemental

oxygen, and maintaining a systolic BP of at least 90 mmHg, as well as continuous fetal monitoring, are critically important [13]. (See "Anaphylaxis during pregnancy, delivery, and lactation", section on 'Management'.)

Laboratory testing — Draw blood in a serum separator tube (ideally within three hours after onset of symptoms) for a tryptase concentration. The result will not be available for hours or days and thus will not aid in or alter the acute management, but is useful during follow-up evaluation [32]. (See "Anaphylaxis: Acute diagnosis", section on 'Laboratory tests'.)

PHARMACOLOGIC TREATMENTS

The tables provide rapid overviews of the emergency management of anaphylaxis in adults (table 1) and children (table 2). The treatment recommendations in this section are consistent with available practice parameters [10-13,32,33]. Each pharmacologic therapy is discussed further below.

Epinephrine — Epinephrine is the first and most important treatment for anaphylaxis, and it should be administered as soon as anaphylaxis is recognized to prevent the progression to life-threatening symptoms (table 1 and table 2). Delayed epinephrine injection is associated with increased fatalities [23-27,34]. Epinephrine should also be administered to patients who have symptoms or signs consistent with impending anaphylaxis when the clinical suspicion for anaphylaxis is high, even if formal diagnostic criteria are not met.

Dosing and administration — Epinephrine is commercially available in different concentrations: 1 mg/mL and 0.1 mg/mL. It is essential to exercise extra caution to ensure the correct dilution is used, as errors can lead to inadvertent overdose, which carries a risk of serious cardiovascular complications and even death [8-10,12,13].

Intramuscular epinephrine injection (preferred) — Intramuscular (IM) injection is the preferred route for initial administration of epinephrine for anaphylaxis in most settings and in patients of all ages (table 6) [35,36]. IM injection is recommended over subcutaneous injection because it consistently provides a more rapid increase in the plasma and tissue concentrations of epinephrine [5,6]. IM injection is also preferred over intravenous (IV) bolus because it is faster and is safer (ie, lower risk of cardiovascular complications, such as severe hypertension and ventricular arrhythmias) [35,36].

The epinephrine dilution for IM injection contains 1 mg/mL. Ampules of 1 mg/mL were previously labeled as 1:1000, but to help prevent medication errors, ratio expressions were removed from epinephrine labels in the United States in 2016 [37]. Only the amount per mL is listed [38]. Ratio labeling remains in place in some other countries (eg, Australia).

- **Initial IM dosing** – For situations where an exact dose can be drawn up and administered, the recommended dose of epinephrine for patients of any age is 0.01 mg/kg (maximum dose of 0.5 mg) per single dose, injected IM into the mid-outer thigh (vastus lateralis muscle). The dose should be drawn up using a 1 mL syringe using the 1 mg/mL formulation of epinephrine.

  If an autoinjector is available or estimating the dose helps to avoid delays, the following approach may be used (table 6):

  Weight-based intramuscular epinephrine dosing and administration for anaphylaxis in health care settings

  In patients with severe symptoms or who are rapidly deteriorating, use an autoinjector if drawing up the dose will cause a significant delay.

  There are 3 available autoinjector strengths: 0.1 mg (Auvi-q), 0.15 mg (EpiPen Jr, others), 0.3 mg (EpiPen, others)

Weight	Preferred	Alternative
<10 kg (infants)	Draw up 0.01 mg/kg (0.01 mL/kg of epinephrine 1 mg/mL)	0.1 mg autoinjector If not available, 0.15 mg autoinjector may be given **or** draw up 0.1 mg (0.1 mL of epinephrine 1 mg/mL)
10 to 25 kg (infants and children)	0.15 mg autoinjector	Draw up 0.15 mg (0.15 mL of epinephrine 1 mg/mL)
>25 to 50 kg	0.3 mg autoinjector	Draw up 0.3 mg (0.3 mL of epinephrine 1 mg/mL)
>50 kg	Draw up 0.5 mg (0.5 mL of epinephrine 1 mg/mL)	0.3 mg autoinjector

  Weight-based intramuscular epinephrine dosing and administration for anaphylaxis in health care settings

  Table 6 - larger image below

  - In infants and young children, the leg must be secured to prevent loss of dose or laceration caused by sudden movement of the extremity [39].
    - Infants weighing <10 kg should be given an exact weight-based dose (not estimated) whenever possible. However, if drawing up an exact dose is likely to cause a significant delay in a patient with severe symptoms or who is rapidly deteriorating, the 0.1 mg dose can be given by autoinjector or by drawing up 0.1 mL of the 1 mg/mL solution. If the 0.1 mg autoinjector is not available, the 0.15 mg autoinjector can be used. It is expected that the side effects of epinephrine would be mild and transient at the plasma concentrations achieved [40].
    - Infants and children weighing from 10 to 25 kg can be given 0.15 mg by autoinjector or by drawing up 0.15 mL of the 1 mg/mL solution. We prefer use of the autoinjector in these patients for speed, reliability, and ease of use [41].
  - Patients weighing >25 to 50 kg can be given 0.3 mg by autoinjector or by drawing up 0.3 mL of the 1 mg/mL solution. We prefer use of the autoinjector in these patients for speed, reliability, and ease of use [41].
  - Patients who weigh >50 kg can be given 0.5 mg (equal to 0.5 mL of the 1 mg/mL solution). If the patient is obese, this can be administered using a 1.5-inch needle to penetrate the subcutaneous fat. However, if drawing up an exact dose is likely to cause a significant delay in a rapidly deteriorating patient, the 0.3 mg dose can be given by autoinjector.

  The needle used in adults and children should be long enough to penetrate the subcutaneous adipose tissue over the vastus lateralis muscle. Realistically, however,

IM injection into the thigh may be impossible in some patients, especially those who are overweight or obese [42,43]. Although the best approach in this situation has not been studied, we suggest as deep an injection as possible into the muscle.

- **Assessing response and redosing —** Approximately 90 percent of patients respond to a single dose of IM epinephrine, particularly if it is given promptly after the onset of symptoms [44].

  IM epinephrine may be repeated at five-minute intervals if there is no response or an inadequate response, or even sooner if clinically indicated [4,8,9,11,13,23,45,46]. When additional IM doses are required, typically one or rarely two additional doses are needed [47,48]. Patients with a history of previous anaphylaxis and those presenting with flushing, diaphoresis, or dyspnea were more likely to require multiple doses of epinephrine to control symptoms in an observational study [49].

  In patients who continue to be hypotensive after initial IM epinephrine, IV fluids should be administered. It is also prudent to begin preparing an epinephrine solution for continuous infusion early so that it is ready in case the patient does not respond to repeated IM epinephrine and IV fluids. (See 'Intravenous fluids' above and 'IV epinephrine continuous infusion' below.)

IV epinephrine by slow bolus (use with caution) — Intravenous (IV) bolus epinephrine is associated with significantly more dosing errors and cardiovascular complications than IM epinephrine and should be used only in the setting of immediate life threat (ie, imminent circulatory arrest) when an IV infusion is not yet available [23,26,35,36,50]. Slow, continuous infusion is preferred if patients have not responded to IM injections [8,51]. In an observational study of 362 doses of epinephrine administered to 301 patients for the emergency management of anaphylaxis, there were four overdoses, all of which occurred with IV bolus administration [35]. Adverse cardiovascular events were significantly more likely with IV bolus compared with IM administration (3 of 30 versus 4 of 316, respectively).

Nonetheless, there may be situations in which a slow IV bolus of epinephrine is indicated, such as when a patient is suffering cardiovascular collapse or impending cardiovascular collapse that is refractory to IM epinephrine and volume resuscitation, and an epinephrine infusion is not available. In most cases, an IV infusion can be prepared in under one minute and is safer than bolus dosing. (See 'IV epinephrine continuous infusion' below.)

- In such cases in an adult or adolescent, this is accomplished by the slow administration of a 50 to 100 mcg (0.05 to 0.1 mg) IV bolus of epinephrine, ideally with hemodynamic monitoring. This is best administered by slow push of a diluted 0.1 mg/mL epinephrine solution (also called "cardiac" epinephrine containing 1 mg of epinephrine in 10 mL

prefilled syringes stocked on resuscitation carts) that has been further diluted in normal saline (table 7). To create this solution, 1 mL of the 0.1 mg/mL "cardiac" epinephrine should be diluted to a total of 10 mL, and then 5 to 10 mL of the diluted solution slowly administered in 1 to 2 mL aliquots every 30 to 60 seconds over one to two minutes. **Note that, for anaphylaxis, the dose is one-tenth or less of the IV epinephrine dose used in cardiac arrest (ie, a maximum one-tenth of the total volume of the stocked prefilled syringe is administered).** Bolus administration is followed by at least three minutes of observation before considering repeat dosing. Usually, a response is observed after a single dose, providing sufficient time to prepare an infusion. If the patient remains severely hypotensive or has shown little response in either heart rate or blood pressure (BP) to the first dose, a second dose is administered in the same way. As soon as infusion is available, bolus injection is discontinued and replaced by titration of the solution [11,52,53]. Note that the "cardiac" epinephrine solution was previously labeled 1:10,000, although this practice was discontinued in 2016 in the United States [37]. Ratio labeling remains in place in some other countries (eg, Australia).

- We **avoid** the use of IV epinephrine boluses in infants and children because data are sparse on the efficacy of safety of this approach and dosing is not well established. Children who are not responding to initial IM epinephrine and fluid resuscitation should be treated with an IV infusion of epinephrine.

IV epinephrine continuous infusion — Patients who do not respond to several IM injections of epinephrine **and** aggressive fluid resuscitation should receive epinephrine by intravenous (IV) infusion, with the rate titrated according to response and the presence of continuous hemodynamic monitoring.

- **Preparing infusion solutions** – Epinephrine is commercially available in several dilutions. Use of the correct dilution is essential to avoid overdose [50]. To prepare an epinephrine IV maintenance infusion, the commercially available epinephrine solution (eg, ampule, syringe) must be further diluted. To reduce the risk of making a medication error, we suggest that centers have a protocol available that includes steps on how to prepare and administer an epinephrine infusion in a limited number of standard concentrations. A commercially prepared (premixed) epinephrine infusion solution is becoming available in some areas with a 24-month shelf-life [54,55].

  - **Routine (1 mcg/mL)** – A simple method for quickly preparing a solution of 1 mcg/mL for adults and adolescents is to add 1 mL of 1 mg/mL epinephrine **or** the entire 10 mL contents of a 0.1 mg/mL prefilled "cardiac" epinephrine syringe (a total of 1 mg) to a 1000 mL (1 liter) bag of normal saline. The resultant solution of approximately 1 mcg/mL delivers 1 mcg/minute of epinephrine for each 60 mL/hour of solution infused. Therefore, 120 mL/hour will deliver 2 mcg/minute and so forth (table 8).

- **Concentrated (4 mcg/mL)** – For adolescent/adult patients who have already received large quantities of IV fluids (ie, 4 or more liters), a more concentrated solution (eg, 4 mcg/mL) is preferable. Using a more concentrated solution allows titration of epinephrine infusion and administration of bolus crystalloid solution to be done independent of one another. To prepare a 4 mcg/mL solution, add 1 mL of 1 mg/mL epinephrine **or** the entire 10 mL contents of one 0.1 mg/mL epinephrine syringe to a 250 mL bag of normal saline. The resultant solution of approximately 4 mcg/mL delivers 1 mcg/minute of epinephrine for each 15 mL/hour of infusion. Therefore, 30 mL/hour delivers 2 mcg/minute, 45 mL/hour delivers 3 mcg/minute, and so forth (table 9).

- **Infants and young children (10 mcg/mL)** – For infants and young children, an even more concentrated solution of 10 mcg/mL is more appropriate to avoid excessively large infusion volumes. To prepare a 10 mcg/mL solution, add 1 mL of 1 mg/mL epinephrine **or** the entire 10 mL contents of one 0.1 mg/mL epinephrine syringe to a 100 mL bag of normal saline. The resultant solution of approximately 10 mcg/mL delivers 1 mcg/minute of epinephrine for each 6 mL/hour of solution infused. Therefore, 12 mL/hour will deliver 2 mcg/minute and so forth (table 10 and table 11).

- **Initial infusion rates**:

  - **Adults** – Start the IV epinephrine infusion at **0.1 mcg/kg/minute** (range: 0.05 to 0.2 mcg/kg/minute) and increase it every two to three minutes by 0.05 mcg/kg/minute until BP and perfusion improve. In monitored patients with impending cardiovascular collapse or progression to shock, a higher initial dose of 0.2 mcg/kg/minute and more rapid titration may be warranted. The maximum dose is not known and will be different for every patient, but rarely will a patient require a dose exceeding 1 mcg/kg/minute. If there is no improvement with doses this large, the patient likely has refractory anaphylaxis, and other agents should be administered. (See 'Refractory anaphylaxis' below.)

    As an example, for an 80 kg patient, the starting dose is 8 mcg/minute (calculation: 80 kg x 0.1 mcg/kg/minute). Using the dilute (ie, 1 mcg/mL) solution, the infusion pump is set to deliver 8 mL/minute (equal to 480 mL/hour) (table 8). Or, if using the more concentrated 4 mcg/mL solution, the infusion pump is set to initially deliver 2 mL/minute (equal to 120 mL/hour) (table 9).

    The infusion rate is increased by one-half of the starting rate every few minutes (in this example, increase infusion by 4 mL/minute [240 mL/hour] if using the dilute 1 mcg/mL solution **or** by 1 mL/minute [60 mL/hour] if using the more concentrated 4 mcg/mL solution). The dose is titrated to effect on BP with continuous noninvasive

monitoring. When BP increases by 10 to 15 percent, the infusion rate is held at that level for three to five minutes to see if improvement continues, and the infusion rate is then adjusted accordingly [12]. If there is a risk of volume overload, the more dilute solution should only be used as a temporizing measure until a more concentrated solution is available. To reduce the risk of making a medication error, we suggest that centers have a protocol available that includes steps on how to prepare and administer epinephrine infusion.

- **Infants and children** – The dose for IV infusion of epinephrine is **0.1 to 1 mcg/kg/minute** with use of an infusion pump, titrated to effect on BP with continuous cardiac monitoring and frequent noninvasive BP monitoring. To reduce the risk of making a medication error, we suggest that centers have a protocol available that includes steps on how to prepare and administer epinephrine infusion. Examples of pediatric infusions are provided (table 10 and table 11).

Epinephrine can be administered through an appropriately positioned, large-bore, peripheral venous catheter with close monitoring of the catheter site throughout the infusion to avoid extravasation injury. The decision to establish central venous access for the epinephrine infusion is discussed separately. (See "Use of vasopressors and inotropes", section on 'Administration and vascular access'.)

Epinephrine nasal spray — Epinephrine nasal spray was approved for the treatment of allergic reactions, including anaphylaxis, by the US Food and Drug Administration in 2024. Each single-use device contains 1 or 2 mg per 0.1 mL. The dose is 1 mg for those weighing 15 kg to less than 30 kg and 2 mg for those weighing 30 kg or greater and is only for intranasal use. The dose is administered into one nostril and can be repeated in the same nostril after five minutes as needed. The pharmacokinetics and pharmacodynamics are similar to epinephrine administered by the non-autoinjector (ie, manual syringe) IM route, although data in patients actually experiencing anaphylaxis are limited. For this reason, we prefer IM administration of epinephrine in most situations.

Mechanisms of action — The pharmacologic actions of epinephrine address the pathophysiologic changes that occur in anaphylaxis better than any other medication. It decreases mediator release from mast cells [56], prevents or reverses obstruction to airflow in the upper and lower respiratory tracts, and prevents or reverses cardiovascular collapse (table 1 and table 2).

The therapeutic actions of epinephrine include the following (table 12) [4,23-27,45]:

- **Alpha-1-adrenergic agonist effects** – Vasoconstriction, increased peripheral vascular resistance, and decreased mucosal edema (eg, in the upper airway).

- **Beta-1-adrenergic agonist effects** – Increased inotropy and increased chronotropy. Although epinephrine increases heart rate in healthy individuals, it often leads to a paradoxical decrease (to normal) in the reflex tachycardia characteristic of anaphylaxis.

- **Beta-2-adrenergic agonist effects** – Bronchodilation and decreased release of mediators of inflammation from mast cells and basophils.

Adverse effects — In patients of all ages, epinephrine administered in therapeutic doses by any route often causes mild, transient pharmacologic effects, such as anxiety, restlessness, headache, dizziness, palpitations, pallor, and tremor [4,26,27,45]. These symptoms and signs are similar to those occurring during the physiologic "fight-or-flight" response due to endogenous epinephrine that occurs normally in sudden frightening or life-threatening situations.

The small risk of an epinephrine-related complication must be weighed against the greater risk of complications associated with severe anaphylaxis. Rarely, epinephrine may lead to ventricular arrhythmias, myocardial ischemia and/or infarction, pulmonary edema, sudden sharp increase in BP, and intracranial hemorrhage. However, anaphylaxis itself can lead to myocardial ischemia and infarction and cardiac arrhythmias in the absence of any exogenous epinephrine or before exogenous epinephrine is administered [57].

Serious adverse effects occur most commonly after an IV bolus injection, particularly if an inappropriately large dose is administered [23,26,35,50]. (See 'Situations requiring caution' below.)

Situations requiring caution — There are **no** absolute contraindications to epinephrine use in anaphylaxis [8-10,12,13,23-27,45]. The risk of death or serious neurologic sequelae from hypoxic-ischemic encephalopathy due to inadequately treated anaphylaxis outweighs other concerns [8-10,12,13,23-27,45]. Existing evidence clearly favors the benefit of epinephrine administration in anaphylaxis. However, formal risk-benefit analyses are not possible, and clinical judgment is essential.

Patients with anaphylaxis and cardiovascular diseases should receive epinephrine, preferably IM, even though they might theoretically be at higher risk for adverse effects. The small number of studies examining epinephrine use in older adults with anaphylaxis suggests that, while IV epinephrine is associated with more adverse cardiovascular effects, IM injection is well tolerated [35,36]. Reluctance to administer epinephrine due to fear of adverse cardiac effects should be countered by the awareness that the heart is a target organ in anaphylaxis. In the healthy human heart, mast cells are present throughout the myocardium and in the intima of coronary arteries. In patients with coronary artery disease, mast cells are found in atherosclerotic lesions and contribute to atherogenesis. Anaphylaxis can unmask subclinical coronary artery disease, and myocardial infarction and/or arrhythmias can occur during anaphylaxis, even if epinephrine is not injected [58,59].

Moreover, anaphylaxis itself can cause vasospasm, arrhythmias, and myocardial infarction in patients, including children, with healthy hearts as confirmed by normal electrocardiograms, echocardiograms, and coronary angiograms after resolution of anaphylaxis [60].

Patients with anaphylaxis taking monoamine oxidase inhibitors (which block epinephrine metabolism), tricyclic antidepressants (which prolong epinephrine duration of action), or stimulants (eg, amphetamines, methylphenidate, cocaine) or those with certain preexisting conditions, such as recent intracranial surgery, aortic aneurysm, or uncontrolled hyperthyroidism or hypertension, should receive epinephrine, preferably IM, even though they theoretically might be at higher risk for adverse effects.

Efficacy — Epinephrine is the best-studied medication in anaphylaxis, although randomized, placebo-controlled trials of epinephrine in humans experiencing anaphylaxis have never been performed for ethical reasons. The evidence for its use comes from observational studies; randomized, controlled clinical pharmacology studies in patients not experiencing anaphylaxis; studies of anaphylaxis in animal models; and epidemiologic studies, including fatality studies. Several case series have implicated the failure to administer epinephrine early in the course of treatment as a consistent finding in anaphylaxis deaths [14-20,61].

- In a series of 13 fatal and near-fatal food-induced anaphylactic reactions in children and adolescents, the six patients who died had symptom onset within one to five minutes after ingestion of the culprit food but did not receive their first dose of epinephrine until between 25 and 180 minutes after the food was ingested. The authors concluded that the failure to recognize severity of the reactions and to administer epinephrine promptly increased the risk of a fatal outcome [15].

- In a series of anaphylactic deaths occurring from 1992 to 1998, only 20 percent of 24 patients were given epinephrine at any point in their treatment [16].

- In the fatality series described previously, only 14 percent of the 164 patients dying from anaphylaxis received epinephrine before respiratory or cardiac arrest, although 62 percent of the 164 patients eventually received it before demise [14].

In addition, anaphylaxis occurring during evaluation of venom immunotherapy has been investigated prospectively. In one study, 68 patients with a history of anaphylaxis to insect stings were randomly assigned to venom immunotherapy or placebo immunotherapy [3]. Following this, all were stung in a controlled, monitored setting and treated (if needed) with a standardized protocol of high-flow oxygen, epinephrine infusion, and normal saline rapid infusion. Nineteen of the 21 patients in the placebo group developed anaphylaxis and received epinephrine. Symptoms responded within five minutes in all but one patient. In

nine patients, an initial attempt to stop the epinephrine infusion was followed by a return of symptoms, which subsided again once the epinephrine infusion was restarted.

In a retrospective chart review of 234 children who received epinephrine for food-induced anaphylaxis, treatment with epinephrine prior to arrival to the emergency department was associated with a significantly lower risk of hospitalization [62]. Although the time between food exposure and administration of epinephrine could not be precisely determined, children who received epinephrine earlier (often because they had an epinephrine autoinjector) were released from the emergency department sooner and were less likely to require hospital admission compared with those who received it only after arrival (17 versus 43 percent, respectively).

Finally, there is extensive clinical experience among allergy practitioners with giving epinephrine to treat anaphylaxis occurring in response to immunotherapy. This is a unique situation because allergy clinic staff observe patients closely for the symptoms and signs of anaphylaxis and reactions are detected at very early stages. Over the past few decades, consensus has been reached that even mild systemic reactions are best treated immediately with epinephrine as this appears to prevent progression to more severe symptoms more effectively than any other available therapies. As a result, successive guidelines for treatment of immunotherapy reactions have called for epinephrine to be given as soon as a systemic reaction of **any** severity is detected [63]. In studies in which all or most patients who developed anaphylaxis after allergen immunotherapy were treated promptly with epinephrine injections, symptoms were mild, and no additional injections of epinephrine were given, even in the 10 to 23 percent of reactions that were biphasic [64].

Glucagon for patients taking beta-blockers — Beta blocker therapy, alone or in combination with angiotensin-converting enzyme inhibitors, has been associated with more severe anaphylaxis [65]. However, in a retrospective study of 789 patients presenting to the emergency department, patients taking beta blockers were no more likely to need epinephrine or to need additional doses of epinephrine than patients not taking beta blockers, so the clinical impact may be relatively small [66]. Epinephrine should still be used as first-line therapy, even in patients taking beta blockers. If a patient taking beta blockers is refractory to epinephrine, glucagon can provide inotropic and chronotropic effects that are not mediated through beta receptors [67,68].

- Adult dosing is 1 to 5 mg slow IV bolus over five minutes. This may be followed by an infusion of 5 to 15 mcg/minute titrated to effect.
- Pediatric dosing is 20 to 30 mcg/kg (maximum 1 mg) slow IV bolus over five minutes. This may be followed by an infusion of 5 to 15 mcg/minute titrated to effect (ie, not weight based).

Rapid administration of glucagon can induce vomiting, so it should be used cautiously in patients with an unprotected airway.

Adjunctive agents — Epinephrine is first-line treatment for anaphylaxis, and there is no known equivalent substitute. H1 and H2 antihistamines, bronchodilators, and glucocorticoids are second-line, adjunctive therapies for anaphylaxis and should only be used for residual symptoms after adequate epinephrine treatment [10,12,13,33,34,69]. There is no direct evidence supporting the benefit of antihistamines or glucocorticoids in anaphylaxis, although they remain in common use, and they should not be considered part of first-line therapy. Patients treated with epinephrine generally do not need any additional adjunctive agents. If symptoms or signs persist, more epinephrine should be administered. When a patient's anaphylaxis manifestations have cleared after epinephrine but the patient continues to report itching or urticaria, an antihistamine may be helpful in relieving the symptom. Antihistamines **do not** relieve upper or lower airway obstruction or shock.

H1 antihistamines — For residual itching or urticaria in patients with anaphylaxis who have already responded adequately to epinephrine, we suggest administering a second-generation H1 antihistamine.

Second-generation H1 antihistamines (eg, cetirizine) have the advantage of being less sedating than first-generation agents (eg, diphenhydramine, hydroxyzine) [27,34,69]. In addition, some first-generation antihistamines (eg, promethazine) can precipitate hypotension if given by rapid IV bolus [70]. Cetirizine is available in both IV and oral formulations, while other nonsedating antihistamines (eg, levocetirizine, loratadine, desloratadine, fexofenadine) are only available in oral forms (table 13). For anaphylaxis, the IV route is preferred initially because gut mucosal edema may not be fully resolved. If IV access is not yet established and the patient is conscious and not at risk for aspiration, oral administration can be considered.

A systematic review of the literature failed to retrieve any randomized, controlled trials that support the use of H1 antihistamines in anaphylaxis [34]. However, H1 antihistamines are effective in acute urticaria, as described elsewhere. (See "New-onset urticaria (hives)", section on 'H1 antihistamine therapy for all patients'.)

Examples of commonly used H1 antihistamines include the following (table 13):

- **Cetirizine** – For adults and children ≥12 years of age, cetirizine 10 mg can be administered IV over one to two minutes. The duration of action approaches 24 hours.

  Children 6 to 11 years of age can receive 5 or 10 mg IV. Children aged six months to five years can receive 2.5 mg IV.

- **Diphenhydramine** – For adults and children ≥12 years of age, diphenhydramine 25 to 50 mg can be administered IV over five minutes, which may be repeated every four to six hours if necessary up to a maximum daily dose of 400 mg per 24 hours. Excessively rapid IV administration may increase hypotension [71]. Diphenhydramine may cause significant sedation.

  For children weighing less than 50 kg, diphenhydramine 1 mg/kg (maximum 50 mg) can be administered IV over five minutes, which may be repeated up to a maximum daily dose of 5 mg/kg or 200 mg per 24 hours.

H2 antihistamines — For residual itching or urticaria in patients with anaphylaxis who have already responded adequately to epinephrine, some contributors to this topic will also administer an H2 antihistamine in addition to the H1 antihistamine. A reasonable option is famotidine (adult dose 20 mg, pediatric dose 0.25 mg/kg to a maximum of 20 mg/dose) IV infused over at least two minutes. Two small, randomized trials have found that adding an IV H2 antihistamine to an IV H1 antihistamine may lead to more rapid symptom resolution, although these studies had methodologic limitations [72,73]. If given, a single IV dose is generally adequate. Patients do not need to be discharged with a course of oral H2 antihistamines.

Bronchodilators — For the treatment of bronchospasm not responsive to epinephrine, inhaled bronchodilators (eg, albuterol, salbutamol) should be administered. Bronchodilators are an adjunctive treatment to epinephrine because they do not prevent or relieve mucosal edema in the upper airway or shock, for which the alpha-1-adrenergic effects of epinephrine are required [8-10,12,13]. The evidence for the use of beta-2-adrenergic agonists in anaphylaxis is extrapolated from their use in acute asthma. (See "Acute exacerbations of asthma in adults: Emergency department and inpatient management", section on 'Inhaled beta-agonists'.)

Glucocorticoids — Our approach is to **not** administer glucocorticoids routinely to patients who respond well to one or two doses of epinephrine. In patients with anaphylaxis who respond promptly to epinephrine, there is no convincing evidence that glucocorticoids provide additional benefit [74,75]. In the past, glucocorticoids were believed to prevent biphasic reactions and given primarily for this reason. However, a 2020 systematic review failed to find evidence for this effect [12]. (See "Biphasic and protracted anaphylaxis", section on 'Prevention'.)

For patients with persistent bronchospasm (especially with known asthma) despite an otherwise adequate response to epinephrine, it is reasonable to administer a glucocorticoid. It is also reasonable to administer glucocorticoids for patients with severe symptoms requiring more than two doses of IM epinephrine or requiring IV epinephrine. Options include methylprednisolone 80 to 125 mg in adults, 1 to 2 mg/kg/dose, maximum

dose 125 mg in children; prednisone 40 to 60 mg orally in adults; or prednisolone 1 to 2 mg/kg, maximum dose 60 mg, in children.

If glucocorticoids are given as adjunctive therapy, they do not necessarily need to be continued as an outpatient taper.

REFRACTORY ANAPHYLAXIS

Definition — Refractory anaphylaxis has historically had variable definitions. In 2020, a multidisciplinary panel proposed a definition that was adopted by the Joint Task Force on Practice Parameters [76,77]. Refractory anaphylaxis is highly likely when **both** of the following criteria are fulfilled:

- Anaphylaxis is present despite following appropriate epinephrine dosing and symptom-directed medical management (eg, intravenous [IV] fluid bolus for hypotension).
- The initial reaction has been treated with three or more appropriate doses of epinephrine (or initiation of an IV epinephrine infusion).

In a study of the European Anaphylaxis Registry, refractory anaphylaxis (defined as a lack of response to the equivalent of two or more doses of epinephrine from an autoinjector) was rare, accounting for less than 0.5 percent of all reported anaphylaxis cases [78]. One-half of cases were caused by medications (mostly in the perioperative setting with IV administered drugs), followed by food and insects. Perioperative refractory anaphylaxis has a high mortality rate; the management of perioperative anaphylaxis is discussed separately. (See "Perioperative anaphylaxis: Recognition and emergency management", section on 'Management of suspected anaphylaxis'.)

Therapies — In addition to intramuscular (IM) and/or IV epinephrine, antihistamines, glucocorticoids, IV fluids, and glucagon for those on beta blockers, a patient with refractory anaphylaxis can be treated with some or all of the following. Anaphylactic shock displays features of both distributive (vasodilatory) and hypovolemic shock. There are no published prospective studies on the optimal management of refractory anaphylaxis. None of the patients in the European Anaphylaxis Registry study received the agents discussed here [78].

Other vasopressors — Another vasopressor (eg, vasopressin) can be added if the patient continues to be hypotensive despite maximal IV epinephrine and fluid therapy. Other adrenergic vasopressors can also be tried, such as norepinephrine and dopamine [11]. It is unclear if the addition of other pressors is superior to epinephrine alone, but one theory about the pathogenesis of refractory anaphylaxis proposes that the clinical manifestations may become refractory to further catecholamine administration, perhaps due to saturation or desensitization of adrenergic receptors [79]. The use of vasopressin in the management

of vasodilatory shock in adults is discussed elsewhere. (See "Use of vasopressors and inotropes".)

Methylene blue — Methylene blue, an inhibitor of nitric oxide synthase and guanylate cyclase, can be added for a patient with vasoplegia from refractory anaphylaxis. The efficacy and ideal dose of methylene blue is unknown, but a single bolus of 1 to 2 mg/kg given over 20 to 60 minutes has been used in cardiac surgery. Evidence is based on a few case reports and other publications, mostly in the perioperative setting [80-83]. Improvement of vasoplegia (eg, increased systemic vascular resistance, reduced vasopressor dose) has been observed within one to two hours in the setting of cardiac surgery, but few data are available about anaphylaxis. (See "Postoperative complications among patients undergoing cardiac surgery", section on 'Vasodilatory shock' and "Methemoglobinemia", section on 'Methylene blue (MB)'.)

Extracorporeal membrane oxygenation — Case reports describe patients with refractory anaphylaxis successfully treated with extracorporeal membrane oxygenation (ECMO) (table 14) or operative cardiopulmonary bypass [84-89]. ECMO can provide circulatory support and oxygenation for patients unresponsive to complete resuscitative efforts in institutions with trained teams and equipment. If available, the ECMO team should be consulted early for patients unresponsive to traditional resuscitative measures. (See "Extracorporeal life support in adults in the intensive care unit: Overview".)

DISPOSITION

Duration of observation — There is no consensus regarding the optimal observation period for a patient who has been successfully treated for anaphylaxis (ie, following resolution of symptoms) in a health care facility [12,47,90-97]. Patients with uniphasic anaphylaxis who received epinephrine in a timely manner and responded promptly to a single dose with resolution of symptoms are at low risk for recurrent symptoms, and observation for one hour after the complete resolution of symptoms is generally sufficient.

The major concern about premature discharge is that a biphasic reaction may occur, although available evidence suggests that this develops in only approximately 5 percent of patients [92,97-100]. Recurrent symptoms are usually less severe than the initial symptoms and often include only urticaria, which does not meet criteria for anaphylaxis. In a study of emergency department adult patients with anaphylaxis, a clinically important biphasic reaction only occurred in 0.4 percent (2 out of 496 patients), and there were no fatalities [90]. A retrospective study of 5641 children seen in emergency departments for acute allergic reactions and treated with epinephrine found less than a 2 percent hourly increase in cumulative proportion of needing additional epinephrine after approximately two- and four-hours for those without and with cardiovascular involvement, respectively [100].

Risk factors for biphasic reactions (table 15) are not known with certainty, but several meta-analyses found that patients with a severe initial presentation (such as hypotension or severe respiratory compromise), the need for more than one dose of epinephrine, delayed time to first epinephrine, or unknown trigger may be at increased risk for recurrent symptoms [12,101]. Biphasic anaphylaxis is discussed in more detail separately. (See "Biphasic and protracted anaphylaxis".)

We suggest the following [33]:

- All patients with anaphylaxis should be observed until symptoms have completely resolved, including at least one hour completely symptom free.
- All patients with severe anaphylaxis (presenting with hypotension or hypoxia or requiring more than two doses of epinephrine), risk factors for biphasic reactions (table 15), or a history of severe asthma should be observed for a minimum of 12 hours, including at least 4 hours completely symptom-free in an appropriate observation unit or inpatient hospital setting. This observation period can be customized based on patient characteristics, reaction severity, patient access to and ability to self-administer epinephrine, patient access to emergency medical care, and shared decision-making.

Discharge care — All patients who have experienced anaphylaxis should be sent home with an anaphylaxis emergency action plan, an epinephrine prescription with instructions for use, printed information about anaphylaxis and its treatment, and instructions to follow up with an allergist. All instructions provided to the patient should ideally be documented in the medical records.

Allergen avoidance and general instructions — Educate the patient that they have experienced anaphylaxis or "killer allergy," which is a life-threatening condition, and that they are at risk for repeat episodes of anaphylaxis in the future. Emphasize the importance of avoiding the suspected cause and that it is not unusual if the cause is unclear [102]. The patient, their family, and friends should learn about anaphylaxis, be able to recognize common symptoms and signs, and have an action plan in case of an emergency.

Anaphylaxis emergency action plan — Patients should be given a written, personalized anaphylaxis emergency action plan that lists the common symptoms and signs of anaphylaxis and contains information about prompt recognition of anaphylaxis and self-administration of epinephrine. Advise the patient that symptoms of the current episode may recur (without further exposure to the causal agent) up to three days after the initial onset of symptoms. Most recurrent symptoms are mild and often restricted to isolated urticaria, which can be managed with antihistamines. However, in the unlikely event of recurrence of severe symptoms such as difficulty breathing or lightheadedness, patients should administer epinephrine and return to the emergency department.

Action plan forms are available in English and Spanish from:

- The Food Allergy Research and Education (Food Allergy & Anaphylaxis Emergency Care Plan)
- The American Academy of Pediatrics (Allergy and Anaphylaxis Emergency Plan English and Spanish)

These plans are appropriate for patients with anaphylaxis from any cause.

Self-administered epinephrine — Patients should be instructed in how to use an epinephrine autoinjector or intranasal device correctly, receive a prescription for it, and be advised to fill the prescription immediately (ie, on the way home). Ideally, two epinephrine autoinjectors or intranasal devices should be prescribed [49,103,104]. In the US, all such devices are dispensed only in two packs. Explain the importance of carrying the epinephrine at all times. Recommendations for dosing and prescribing epinephrine for use by patients and caregivers in the community setting are provided separately. (See "Prescribing epinephrine for anaphylaxis self-treatment", section on 'Dosing'.)

Instructions in the proper use of epinephrine autoinjectors and intranasal devices should be reviewed verbally, and patients should be given written material or directed to a manufacturer's website providing relevant information. Advise the patient to make sure that family and friends are aware of the serious nature of anaphylaxis, the causes, and how to administer epinephrine. The correct autoinjector technique is important to avoid unintentional injection into fingers, thumbs, or other body parts. Patient information that can be printed or accessed online should be provided. These steps prior to discharge are often overlooked. In a survey of 1885 patients who survived anaphylaxis, 28 percent of those who did not self-administer epinephrine reported that they had never received an epinephrine prescription [105]. (See 'Information for patients' below.)

Adjunctive medications — Advise the patient to continue to take antihistamines as needed for hives. If glucocorticoids were administered as part of the acute management, it is not necessary to continue as an outpatient taper.

Follow-up care — Advise the patient to follow up with an allergist (with a direct referral if possible or through their primary care clinician) for ongoing management, to identify the cause, and to reduce the risk of recurrence. Specifically, the suspected cause for the anaphylactic episode should be verified. In a study of over 500 emergency department patients with suspected anaphylaxis who followed up with an allergist/immunologist, 35 percent of patients had an alteration in the diagnosis or suspected cause [102]. (See "Anaphylaxis: Confirming the diagnosis and determining the cause(s)" and "Long-term management of patients with anaphylaxis".)

RISK OF RECURRENCE

Patients who have experienced anaphylaxis are at risk for recurrent episodes unless long-term risk reduction measures are implemented. The risk of recurrence has been estimated in several retrospective studies of different patient populations [12,13,106,107]. In a prospective study of nearly 300 children with anaphylaxis requiring acute medical care (predominantly caused by foods), the annual recurrence rate was 18 percent [108]. Concomitant asthma and the need for epinephrine to treat the initial episode further increased the risk of recurrence. Thus, it is important to equip patients with epinephrine autoinjectors or intranasal devices and arrange appropriate referrals in a timely manner after the initial event.

For anaphylaxis caused by stinging insects, a course of immunotherapy can dramatically reduce the risk of a recurrent reaction. (See "Hymenoptera venom immunotherapy: Indications, efficacy, and mechanism of action".)

Comorbidities, such as asthma, other chronic pulmonary disease, and cardiovascular disease, should be optimally controlled as they can increase the risk of fatal anaphylaxis. (See "Fatal anaphylaxis".)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Anaphylaxis".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see "Patient education: Anaphylaxis (The Basics)" and "Patient education: Anaphylaxis – Discharge instructions (The Basics)" and "Patient education: How to give

epinephrine (The Basics)")

- Beyond the Basics topics (see "Patient education: Anaphylaxis symptoms and diagnosis (Beyond the Basics)" and "Patient education: Anaphylaxis treatment and prevention of recurrences (Beyond the Basics)" and "Patient education: Using an epinephrine autoinjector (Beyond the Basics)")

Other sources of accurate patient information accessible through the internet include the American Academy of Allergy, Asthma, and Immunology and the American College of Allergy, Asthma, and Immunology [109,110].

SUMMARY AND RECOMMENDATIONS

- **Act fast** – Patients with anaphylaxis should be assessed and treated as rapidly as possible, as respiratory or cardiac arrest and death can occur within minutes. Anaphylaxis appears to be most responsive to treatment in its early phases (within 60 minutes of onset of symptoms). Management is summarized in rapid overview tables for adults (table 1) and children (table 2). (See 'Initial assessment and management' above.)

- **Epinephrine is lifesaving** – Epinephrine should be injected as soon as anaphylaxis is recognized to prevent progression of the reaction. There are no absolute contraindications to epinephrine use in anaphylaxis of any severity. We recommend epinephrine for patients with apparently mild symptoms and signs (eg, a few hives and mild wheezing) (**Grade 1B**), as well as for patients with moderate-to-severe symptoms and signs (**Grade 1A**). Most patients respond to one or two doses of intramuscular (IM) epinephrine when it is given promptly after symptoms begin. (See 'Epinephrine' above and 'Efficacy' above.)

  - IM injection into the mid-outer thigh is the preferred route for initial administration of epinephrine in most settings and in patients of all ages (table 6). When an exact dose can be drawn up and administered, 0.01 mg/kg (maximum of 0.5 mg) should be administered every five minutes or more frequently, if necessary. (See 'Intramuscular epinephrine injection (preferred)' above.)

  - When an autoinjector is used, children weighing less than 25 kg should receive the 0.15 mg dose, and those weighing over 25 kg should receive the 0.3 mg dose administered every five minutes or more frequently, if necessary.

- **Address airway, breathing, and circulation** – Administer high-flow supplemental oxygen and establish intravenous (IV) access. Tracheal intubation should be performed on an emergency basis if stridor or respiratory distress due to upper airway compromise is present. An emergency cricothyroidotomy may be required to secure the airway if severe upper airway edema prevents access to the glottic aperture, even with use of a video

laryngoscope. Fluid resuscitation with large volumes of crystalloid should be initiated immediately in patients who present with hypotension or incomplete response to IM epinephrine. Massive fluid shifts can occur in anaphylaxis. (See 'Initial stabilization' above.)

- **Patients not responding to IM epinephrine** – IV epinephrine is indicated for patients with persistent hypotension or symptoms and signs suggestive of impending shock who do not respond to one or two initial IM injections of epinephrine and fluid resuscitation. For these patients, we suggest that epinephrine be administered by continuous IV infusion rather than by intermittent slow IV bolus (**Grade 2C**). IV infusion is less likely to cause extreme hypertension or ventricular arrhythmias. Epinephrine infusion should be accompanied by continuous hemodynamic monitoring. (See 'IV epinephrine by slow bolus (use with caution)' above and 'IV epinephrine continuous infusion' above.)

- **Adjunctive agents** – Once epinephrine has been administered, adjunctive therapies may be given for residual symptoms (table 1 and table 2). These agents should not be used instead of epinephrine and should not delay the administration of epinephrine.

  Adjunctive agents include:

  - Antihistamines for persistent itching or urticaria. (See "New-onset urticaria (hives)", section on 'H1 antihistamine therapy for all patients' and 'H2 antihistamines' above.)

  - Bronchodilators and/or glucocorticoids for persistent bronchospasm. (See "Acute exacerbations of asthma in adults: Emergency department and inpatient management", section on 'Inhaled beta-agonists' and "Acute exacerbations of asthma in adults: Emergency department and inpatient management", section on 'Systemic glucocorticoids'.)

  Patients without residual symptoms after epinephrine treatment do not require additional therapy. The available evidence suggests that glucocorticoids or antihistamines do not prevent recurrent symptoms (biphasic anaphylaxis). (See "Biphasic and protracted anaphylaxis", section on 'Prevention'.)

- **Disposition** – All patients with anaphylaxis should be observed until symptoms have completely resolved, including at least one hour completely symptom-free. Patients with uniphasic anaphylaxis who received epinephrine in a timely manner and responded promptly to a single dose with resolution of symptoms are low risk for recurrent symptoms, and observation for one hour after the complete resolution of symptoms is generally sufficient. All patients with severe anaphylaxis (presenting with hypotension or hypoxia or requiring more than two doses of epinephrine), patients with risk factors for biphasic reactions (table 15), or patients with a history of severe asthma should be observed for a minimum of 12 hours, including at least 4 hours completely symptom-free, in an

appropriate observation unit or inpatient hospital setting. (See 'Duration of observation' above.)

- **Follow-up care** – Patients successfully treated for anaphylaxis should be discharged with a personalized, written anaphylaxis emergency action plan; an epinephrine autoinjector or intranasal device; written information about anaphylaxis and its treatment; and referral to an allergist for further evaluation. It is important to make all efforts to confirm the cause to reduce the risk of recurrence. Anaphylaxis action plans for patients of any age (Food Allergy & Anaphylaxis Emergency Care Plan) are available. (See 'Discharge care' above and 'Follow-up care' above and 'Risk of recurrence' above and 'Information for patients' above.)

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Topic 392 Version 72.0

GRAPHICS

Table 1: **Anaphylaxis in adults: Rapid overview of emergency management**

Diagnosis is made clinically:
The most common signs and symptoms are cutaneous (eg, sudden onset of generalized urticaria, angioedema, flushing, pruritus). However, 10 to 20% of patients have no skin findings.
Danger signs – Rapid progression of symptoms, respiratory distress (eg, stridor, wheezing, dyspnea, increased work of breathing, persistent cough, cyanosis), vomiting, abdominal pain, hypotension, dysrhythmia, chest pain, collapse.
Acute management:
The first and most important treatment in anaphylaxis is epinephrine. There are **NO absolute contraindications to epinephrine** in the setting of anaphylaxis. Epinephrine is given immediately upon recognition of anaphylaxis.
Airway – Early intubation if evidence of impending airway obstruction from angioedema. Delay may lead to complete obstruction. Intubation can be difficult and should be performed by the most experienced clinician available. Cricothyrotomy may be necessary.
Monitoring – Continuous noninvasive hemodynamic monitoring and pulse oximetry monitoring should be performed.
Promptly and simultaneously, give:
IM epinephrine (1 mg/mL preparation) – Give epinephrine 0.3 to 0.5 mg IM in the mid-outer thigh. Can repeat every 5 minutes (or more frequently), as needed. If epinephrine is injected promptly IM, most patients respond to 1, 2, or at most, 3 doses. If symptoms are not responding to epinephrine injections, prepare IV epinephrine for infusion (refer to below).
If hypotensive, place patient in recumbent position, if tolerated, and elevate lower extremities.
Oxygen – Give with nonrebreather mask at 15 liters/minute flow rate or commercial high-flow oxygen masks (providing at least 70% and up to 100% oxygen), as needed.
Crystalloid (eg, Ringer's lactate or normal saline) rapid bolus – Establish two large-bore IV lines. Treat hypotension with rapid infusion of 1 to 2 liters IV. Repeat, as needed. Massive fluid shifts with severe loss of intravascular volume can occur. Monitor urine output.
Albuterol – For bronchospasm resistant to IM epinephrine, give 2.5 to 5 mg in 3 mL saline via nebulizer, or 2 to 3 puffs by metered dose inhaler. Repeat, as needed.
Adjunctive therapies (for residual symptoms after adequate response to epinephrine):
H1 antihistamine* **–** For residual itching or urticaria, give cetirizine (preferred) 10 mg IV (given over 2 minutes) or diphenhydramine 25 to 50 mg IV (given over 5 minutes).
H2 antihistamine* **–** For residual itching or urticaria, may give famotidine 20 mg IV (given over 2 minutes).
Glucocorticoid* **–** For residual bronchospasm and for patients requiring more than 2 doses of IM epinephrine or IV epinephrine, give methylprednisolone 80 to 125 mg IV or prednisone 40 to 60 mg orally.

Treatment of refractory symptoms:
Epinephrine infusion¶ **–** For patients with inadequate response to IM epinephrine and IV saline, give epinephrine continuous infusion, beginning at **0.1 microgram/kg/minute** by infusion pump[Δ]. Titrate the dose continuously according to blood pressure, cardiac rate and function, and oxygenation.
Vasopressors¶ **–** Some patients may require a second vasopressor (in addition to epinephrine). All vasopressors should be given by infusion pump, with the doses titrated continuously according to blood pressure and cardiac rate/function and oxygenation monitored by pulse oximetry.
Glucagon – Patients on beta blockers may not respond to epinephrine and can be given glucagon 1 to 5 mg IV over 5 minutes, followed by infusion of 5 to 15 micrograms/minute. Rapid administration of glucagon can cause vomiting.
Methylene blue – Single bolus of 1 to 2 mg/kg given over 20 to 60 minutes may improve vasoplegia.
Extracorporeal membrane oxygenation (ECMO) – Early consultation with ECMO team, if available, for patients unresponsive to complete resuscitative efforts.

Instructions on how to prepare and administer epinephrine for IV continuous infusions are available as separate tables in UpToDate.

IM: intramuscular; IV: intravenous.

* These medications should not be used as initial or sole treatment.

¶ All patients receiving an infusion of epinephrine and another vasopressor require continuous noninvasive monitoring of blood pressure, heart rate and function, and oxygen saturation.

Δ For example, the initial infusion rate for a 70 kg patient would be 7 micrograms/minute. This is consistent with the recommended range for non-weight-based dosing for adults, which is 2 to 10 micrograms/minute. Non-weight-based dosing can be used if the patient's weight is not known and cannot be estimated.

Adapted from: Simons FER. Anaphylaxis. J Allergy Clin Immunol 2010; 125:S161.

Graphic 58346 Version 37.0

*

Table 2: **Anaphylaxis in infants and children: Rapid overview of emergency management**

Diagnosis is made clinically:
The most common signs and symptoms are cutaneous (eg, sudden onset of generalized urticaria, angioedema, flushing, pruritus). However, 10 to 20% of patients have no skin findings.
Danger signs – Rapid progression of symptoms, evidence of respiratory distress (eg, stridor, wheezing, dyspnea, increased work of breathing, retractions, persistent cough, cyanosis), signs of poor perfusion, abdominal pain, vomiting, dysrhythmia, hypotension, collapse.
Monitoring – Continuous noninvasive hemodynamic monitoring and pulse oximetry monitoring should be performed.
Acute management:
The first and most important therapy in anaphylaxis is epinephrine. There are **NO absolute contraindications to epinephrine** in the setting of anaphylaxis. Epinephrine is given immediately upon recognition of anaphylaxis.
Airway – Early intubation if evidence of impending airway obstruction from angioedema. Delay may lead to complete obstruction. Intubation can be difficult and should be performed by the most experienced clinician available. Cricothyrotomy may be necessary.
IM epinephrine (1 mg/mL preparation) – Epinephrine 0.01 mg/kg should be injected IM in the mid-outer thigh. For large children (>50 kg), the maximum is 0.5 mg per dose. If there is no response or the response is inadequate, the injection can be repeated in 5 minutes (or more frequently). If epinephrine is injected promptly IM, patients respond to 1, 2, or, at most, 3 injections. If signs of poor perfusion are present or symptoms are not responding to epinephrine injections, prepare IV epinephrine for infusion (refer to below).
If hypotensive, place patient in recumbent position, if tolerated, and elevate lower extremities.
Oxygen – Give using a nonrebreather mask at 15 liters/minute flow rate or commercial high-flow oxygen masks (providing at least 70% and up to 100% oxygen), as needed.
Crystalloid (eg, Ringer's lactate or normal saline) rapid bolus – Establish 2 large-bore IV lines. Treat poor perfusion with rapid infusion of 20 mL/kg. Reevaluate and repeat fluid boluses (20 mL/kg), as needed. Massive fluid shifts with severe loss of intravascular volume can occur. Monitor urine output.
Albuterol – For bronchospasm resistant to IM epinephrine, give albuterol 2.5 mg inhaled via nebulizer. Dilute in saline if using a concentrated albuterol solution (≥0.5%). Repeat, as needed.
Adjunctive therapies (for residual symptoms after adequate response to epinephrine):
H1 antihistamine – For residual itching or urticaria, give cetirizine (preferred; children aged 6 months to 5 years can receive 2.5 mg IV, those 6 to 11 years of age can receive 5 or 10 mg IV, over 2 minutes) or diphenhydramine 1 mg/kg (maximum 50 mg IV, over 5 minutes).
H2 antihistamine – For residual itching or urticaria, may give famotidine 0.25 mg/kg (maximum 20 mg) IV, over at least 2 minutes.

Glucocorticoid – For residual bronchospasm or for patients requiring more than 2 doses of IM or IV epinephrine, give methylprednisolone 1 to 2 mg/kg (maximum 125 mg) IV or prednisolone 1 to 2 mg/kg, maximum 60 mg orally.
Treatment of refractory symptoms:
Epinephrine infusion¶ – In patients with inadequate response to IM epinephrine and IV saline, give epinephrine continuous infusion at 0.1 to 1 microgram/kg/minute, titrated to effect.
Vasopressors¶ – Patients may require large amounts of IV crystalloid to maintain blood pressure. Some patients may require a second vasopressor (in addition to epinephrine). All vasopressors should be given by infusion pump, with the doses titrated continuously according to blood pressure and cardiac rate/function monitored continuously and oxygenation monitored by pulse oximetry.
Glucagon – Patients on beta blockers may not respond to epinephrine and can be given glucagon 20 to 30 micrograms/kg (maximum 1 mg) IV over 5 minutes. Rapid administration of glucagon can cause vomiting.
Methylene blue – Single bolus of 1 to 2 mg/kg given over 20 to 60 minutes may improve vasoplegia.
Extracorporeal membrane oxygenation (ECMO) – Early consultation with ECMO team, if available, for patients unresponsive to complete resuscitative efforts.

IM: intramuscular; IV: intravenous.

* A child is defined as a prepubertal patient weighing less than 40 kg.

¶ All patients receiving an infusion of epinephrine and/or another vasopressor require continuous noninvasive monitoring of blood pressure, heart rate and function, and oxygen saturation. We suggest that pediatric centers provide instructions for preparation of standard concentrations and also provide charts for established infusion rate for epinephrine and other vasopressors in infants and children.

Graphic 74242 Version 41.0

Table 3: **Diagnostic criteria for anaphylaxis**

Anaphylaxis is highly likely when any ONE of the following three criteria is fulfilled:
1. Acute onset of an illness (minutes to several hours) with involvement of the skin, mucosal tissue, or both (eg, generalized hives, pruritus or flushing, swollen lips-tongue-uvula)
AND AT LEAST ONE OF THE FOLLOWING:
A. Respiratory compromise (eg, dyspnea, wheeze-bronchospasm, stridor, hypoxemia)
B. Reduced BP* or associated symptoms of end-organ dysfunction (eg, hypotonia, collapse, syncope, incontinence)
2. TWO OR MORE OF THE FOLLOWING that occur rapidly *after exposure to a LIKELY allergen for that patient* (minutes to several hours):
A. Involvement of the skin mucosal tissue (eg, generalized hives, itch-flush, swollen lips-tongue-uvula)
B. Respiratory compromise (eg, dyspnea, wheeze-bronchospasm, stridor, hypoxemia)
C. Reduced BP* or associated symptoms (eg, hypotonia, collapse, syncope, incontinence)
D. Persistent gastrointestinal symptoms (eg, crampy abdominal pain, vomiting)
3. Reduced BP* *after exposure to a KNOWN allergen for that patient* (minutes to several hours):
A. Infants and children - Low systolic BP (age-specific)* or greater than 30% decrease in systolic BP
B. Adults - Systolic BP of less than 90 mmHg or greater than 30% decrease from that person's baseline

BP: blood pressure.
* Low systolic blood pressure for children is defined as:
- Less than 70 mmHg from 1 month to 1 year
- Less than (70 mmHg + [2 x age]) from 1 to 10 years
- Less than 90 mmHg from 11 to 17 years

Adapted with permission from: Sampson HA, Munoz-Furlong A, Campbell RL, et al. Second symposium on the definition and management of anaphylaxis: summary report-Second National Institute of Allergy and Infectious Disease/Food Allergy and Anaphylaxis Network symposium. J Allergy Clin Immunol 2006; 117:391. Copyright © 2006 The American Academy of Allergy, Asthma, and Immunology.

Graphic 72225 Version 16.0

Table 4: **Anaphylaxis signs and symptoms**

Signs and symptoms of anaphylaxis
Dermatologic – Pruritus, flushing, urticaria, angioedema
Eyes – Conjunctival injection, lacrimation, pruritus, periorbital edema
Nose – Sneezing, rhinorrhea, nasal congestion
Upper airway – Glossal/pharyngeal edema, metallic taste, hoarseness, stridor, sense of choking
Lower airway – Dyspnea, tachypnea, wheezing, dry and repetitive cough, cyanosis
Gastrointestinal/abdominal – Nausea, vomiting, crampy pain, diarrhea, uterine contractions (females)
Cardiovascular – Hypotension, tachycardia (or sometimes bradycardia), palpitations, lightheadedness, syncope
Neurologic – Sense of impending doom, headache

Graphic 58836 Version 7.0

Table 5: **Differentiating allergic urticaria, anaphylaxis, and common angioedema disorders**

	Anaphylaxis (including generalized allergic reaction)	**Urticaria**	**Angioedema disorders (including ACEI use and rare C1 inhibitor deficiency)**
Onset	Minutes to 1 hour (rarely several hours).	Days to weeks, but can occur in minutes to hours if IgE mediated (eg, food, medications).	Hours, peaking at 24 to 36 hours.
Clinical course	Variable and unpredictable, may progress within minutes to respiratory or cardiovascular compromise and death, may be mild and resolve spontaneously. **Generalized allergic reactions may actually be early anaphylaxis and progress to airway compromise or systemic findings.**	Most episodes of acute urticaria improve within 1 to 2 weeks.	Gradually resolves over 1 to 3 days.
Skin findings	Urticaria, flushing/generalized erythema, pruritus; skin findings can be absent in severe cases.	Intensely pruritic urticaria; flushing/generalized erythema can also occur. One-third of patients have urticaria with angioedema*.	**Not present (this is a key differentiating factor).**
Associated symptoms	Dyspnea, throat tightness, hoarse voice, conjunctival erythema, nasal congestion, sneezing, bronchospasm, cough, chest tightness, hypotension, syncope, altered mental status, crampy abdominal pain, vomiting, uterine cramping.	**Systemic findings typically not present: If they are, treat as anaphylaxis.**	ACEI-associated angioedema – Systemic symptoms typically not present. C1 inhibitor deficiency – Can have crampy abdominal pain/vomiting if bowel wall involvement.
Angioedema locations	Lips, tongue, face, uvula, and/or larynx.	Urticaria can affect skin anywhere on the body.	ACEI-associated angioedema – Commonly affects lips, tongue, or

		When accompanied by angioedema or presenting as isolated angioedema, face, lips, extremities, and/or genitals are often involved.	sometimes face, but other sites (eg, larynx) may rarely be affected. C1 inhibitor deficiency – Can affect larynx, intestinal walls, or skin (although usually not all at once).
Mediators	Multiple – Histamine, tryptase, platelet-activating factor, nitric oxide, arachidonic acid metabolites.	Histamine.	Bradykinin.
Testing	Tryptase.[¶]	Typically not helpful.[Δ]	For ACEI-associated angioedema – None. For suspected C1 inhibitor deficiency – C1 inhibitor level and function and C4.[◇]
First-line treatment	Epinephrine; IV fluids if hypotensive.	Antihistamines.	For all – Airway management as necessary. For ACEI-associated angioedema – Discontinue ACEI. For known or suspected C1 inhibitor deficiency – Multiple possible therapies.[§]
Second-line treatment	Antihistamines, glucocorticoids, bronchodilators.	Glucocorticoids.	

This table summarizes differences among anaphylaxis, urticaria, and angioedema disorders. Refer to UpToDate content on the differential diagnosis of anaphylaxis in adults and children for a more exhaustive discussion.

ACEI: angiotensin-converting enzyme inhibitor; IgE: immunoglobulin E; IV: intravenous.

* When angioedema occurs with urticaria, it is assumed to be histamine mediated.

¶ Refers to UpToDate table on tryptase collection for timing and collection of sample.

Δ If a food is the suspected trigger, blood tests for IgE specific to that food may be sent (eg, IgE to milk, shrimp, etc). Testing for most other triggers (medications, insect venoms, exercise, etc) is best performed in consultation with an allergist as part of follow-up care. Refer to UpToDate algorithm on evaluation of new-onset urticaria.

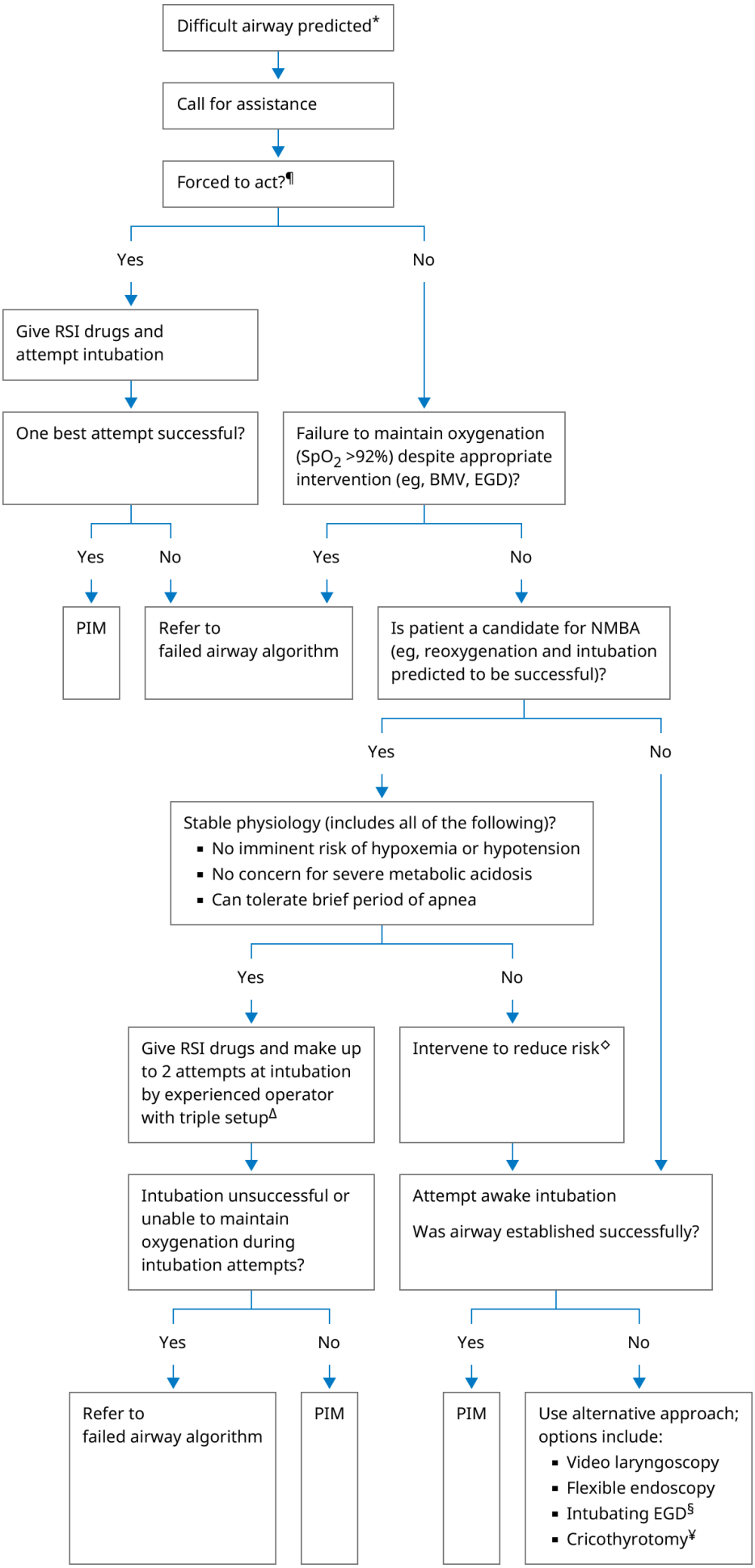
◇ Note that C1 inhibitor deficiency can be hereditary or acquired (usually after age 40 years). A low C4 may be seen in most of these disorders and is a reasonable screening test. However, there are also

angioedema disorders in which complement studies are normal and diagnosis is more complicated. In addition, use of an ACEI can "unmask" these disorders, such that a patient may develop angioedema as a result of both ACEI use and an underlying disorder. For this reason, it is important to refer patients with repeated episodes of angioedema to an allergy/immunology specialist for further evaluation.

§ Hereditary angioedema therapies include sebetralstat, C1 inhibitor concentrate, icatibant, ecallantide, and fresh-frozen plasma.

Graphic 148765 Version 2.0

Algorithm 1: **Difficult emergency airway management algorithm**



BMV: bag-mask ventilation; EGD: extraglottic airway device; NMBA: neuromuscular blocking agent; PIM: post-intubation management; RSI: rapid sequence intubation; SpO_2: oxygen saturation.

* Predicting the difficult airway is discussed in the UpToDate topic and includes assessment tools such as the LEMON mnemonic.

¶ This situation is further discussed in the UpToDate topic and includes, for example, a patient who is agitated and unable to cooperate, or a patient with rapidly progressing airway swelling.

Δ Triple setup implies a primary intubating plan is in place, followed by a nonsurgical rescue device (eg, EGD), and lastly a surgical rescue plan (most often an open surgical bougie-assisted cricothyrotomy).

◇ During RSI, physiologic derangements such as hypotension and hypoxemia increase the risk of cardiovascular collapse during the peri-intubation period. Possible mitigating interventions include the following:

- In the hypotensive patient – An intravenous bolus of isotonic fluid and push-dose vasopressors.
- In the hypoxemic patient – Administration of high-flow oxygen and apneic oxygenation.
- In the patient with severe metabolic acidosis – Treat the underlying etiology and perform awake technique if period of apnea contraindicated.

Refer to UpToDate topics on RSI for further detail.

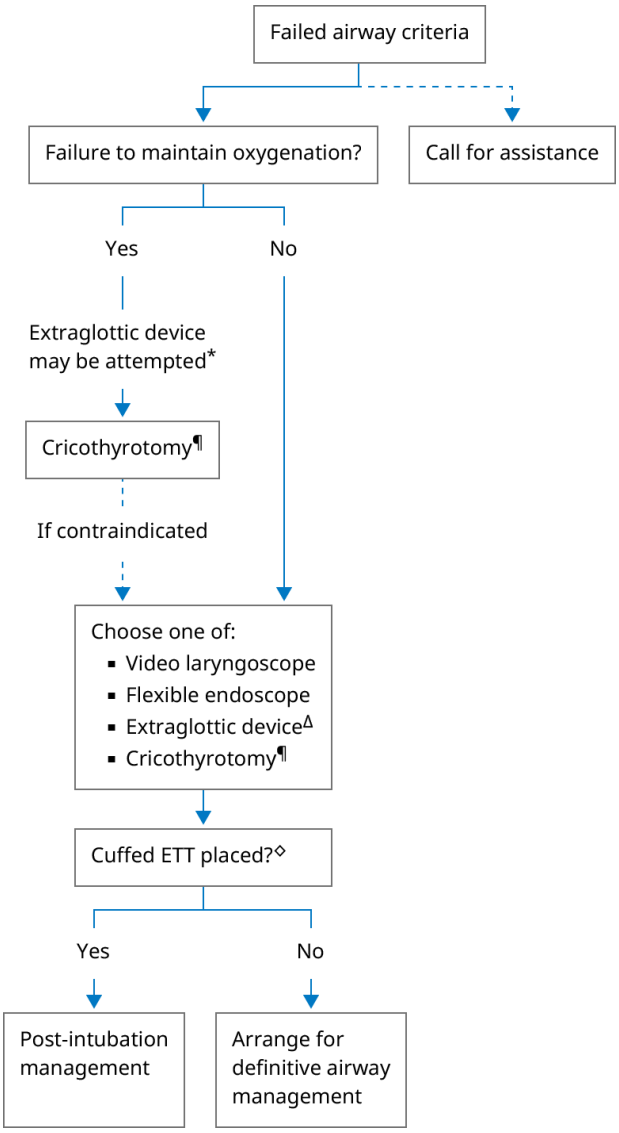
§ The minimum age/weight limit for an intubating EGD is 12 yrs/30 kg.

¥ Needle cricothyrotomy may be performed on patients of any age but is preferable to surgical cricothyrotomy in infants and children up to 10 to 12 years of age because it is easier to perform with less potential damage to the airway. For older patients, surgical cricothyrotomy is typically chosen. For a description of these procedures, refer to UpToDate content on needle and surgical cricothyrotomy.

Adapted from Brown: The Walls Manual of Emergency Airway Management, 6th edition, Philadelphia, Wolters Kluwer, 2022 and The Difficult Airway Course (www.theairwaysite.com). Used with permission.

Graphic 140479 Version 6.0

Algorithm 2: **Failed airway algorithm**



ETT: endotracheal tube.

* Should a surgical airway be needed in these circumstances, one attempt at tracheal intubation using video or direct laryngoscopy may be performed, provided preparations for a surgical airway occur simultaneously.

¶ Needle cricothyrotomy may be performed on patients of any age but is preferable to surgical cricothyrotomy in infants and children up to 10 to 12 years of age because it is easier to perform with less potential damage to the airway. For older patients, surgical cricothyrotomy is typically chosen. For a description of these procedures, refer to UpToDate content on needle and surgical cricothyrotomy.

∆ Placement of an extraglottic device is contraindicated in the setting of severe hypopharyngeal pathology, such as epiglottitis.

◇ In neonates, a cuffed ETT is suggested, but an uncuffed endotracheal tube is acceptable.

From Brown: The Walls Manual of Emergency Airway Management, 6th edition, Philadelphia, Wolters Kluwer, 2022 and The Difficult Airway Course (www.theairwaysite.com). Used with permission.

Graphic 73112 Version 28.0

Table 6: **Weight-based intramuscular epinephrine dosing and administration for anaphylaxis in health care settings**

In patients with severe symptoms or who are rapidly deteriorating, use an autoinjector if drawing up the dose will cause a significant delay.

There are 3 available autoinjector strengths: 0.1 mg (Auvi-q), 0.15 mg (EpiPen Jr, others), 0.3 mg (EpiPen, others)

Weight	Preferred	Alternative
<10 kg (infants)	Draw up 0.01 mg/kg (0.01 mL/kg of epinephrine 1 mg/mL)	0.1 mg autoinjector If not available, 0.15 mg autoinjector may be given **or** draw up 0.1 mg (0.1 mL of epinephrine 1 mg/mL)
10 to 25 kg (infants and children)	0.15 mg autoinjector	Draw up 0.15 mg (0.15 mL of epinephrine 1 mg/mL)
>25 to 50 kg	0.3 mg autoinjector	Draw up 0.3 mg (0.3 mL of epinephrine 1 mg/mL)
>50 kg	Draw up 0.5 mg (0.5 mL of epinephrine 1 mg/mL)	0.3 mg autoinjector

- If the dose is to be drawn up, confirm that the **1 mg/mL** epinephrine solution is being used. Draw up into a 1 mL syringe. The intramuscular injection is given into the mid-outer thigh.
- If using an autoinjector, it should be held in place for 3 seconds after the injection, which is sufficient to deliver the dose.
- Most patients respond to a single dose of intramuscular epinephrine. However, if there is no response or an inadequate response, then intramuscular epinephrine may be repeated at 5- to 15-minute intervals or sooner, if clinically indicated.

Data From:

1. Sicherer SH, Simons FER. Epinephrine for first-aid management of anaphylaxis. Pediatrics 2017; 139:e20164006.

Graphic 118924 Version 2.0

Table 7: **Epinephrine intravenous (IV) slow bolus for treatment of anaphylaxis refractory to intramuscular (IM) epinephrine – Adult or adolescent patient**

Prepare the epinephrine solution: ▪ Use a prefilled **0.1 mg/mL** epinephrine 10 mL syringe (also called "cardiac" or "ACLS" epinephrine). ▪ Take 1 mL of this solution and dilute it with 9 mL of normal saline to make a total volume of 10 mL (final concentration 0.01 mg/mL).
Administer the diluted solution (use with caution): ▪ Slowly administer 5 to 10 mL of the diluted (0.01 mg/mL) solution in 1 to 2 mL aliquots over 1 to 2 minutes while monitoring the patient's heart rate and blood pressure. ▪ Wait at least 3 minutes after administration to assess response. ▪ If the patient is still severely hypotensive or shows little improvement, a second dose can be administered.
Slow IV bolus epinephrine may be used (infrequently) as a temporizing measure in an emergency while a titratable epinephrine infusion solution is being prepared. As soon as the epinephrine infusion solution is available, bolus administration is discontinued and replaced by titration of the infusion; refer to UpToDate topic on emergency management of anaphylaxis and separately available epinephrine infusion tables.

For anaphylaxis, the dose of epinephrine is one-tenth or less of the IV epinephrine dose used in cardiac arrest (ie, a maximum of one-tenth of the total volume of the stocked prefilled syringe is administered).

ACLS: advanced cardiac life support; IV: intravenous.

Graphic 148766 Version 2.0

Table 8: **Example of epinephrine infusion (1 microgram/mL) - Adult**

Example of preparation of epinephrine infusion for refractory symptoms of anaphylaxis (adult patient) for emergency/critical care units

Final concentration: Epinephrine 1 microgram/mL

Preparation

1. **CHECK** vial strength.

2. Add 1 milligram of epinephrine to 1 liter bag of 0.9% normal saline
   - If you have 0.1 mg/mL strength epinephrine (may also be labeled 1:10,000) use 10 mL to deliver 1 mg to bag*
   - If you have 1 mg/mL strength epinephrine (may also be labeled 1:1000) use 1 mL to deliver 1 mg to bag*

Administration

- Start the epinephrine infusion at **0.1 microgram/kg/minute** using a programmable infusion pump while continuously monitoring the patient's cardiac rhythm and blood pressure (ie, approximately 6 to 10 micrograms/minute in most adults).
- Using a body weight of 80 kg as an example, you can determine the infusion rate to deliver a dose of 0.1 microgram/kg/minute from a 1 microgram/mL bag as follows:
  1. Multiply 0.1 microgram/kg/minute dose by 80 kg body weight to determine dose in micrograms/minute (= 8 micrograms/minute).
  2. Use this table to convert 8 micrograms/minute to infusion rate of 8 mL/minute (= 480 mL/hour).
- Every 2 to 3 minutes, increase the infusion rate by approximately one-half of the starting rate until the blood pressure and perfusion improve. For more detail, refer to the UpToDate topic on emergency treatment of anaphylaxis.
- In morbidly obese adults, we suggest initiating the infusion using a standard, ie, non-weight-based dose (eg, up to 10 micrograms/minute initially) and titrating to effect.

Adult infusion dose	**Administration rate for infusion pump to deliver adult dose shown**	
micrograms per minute	**mL per minute**	**mL per hour**
1	1	60
2	2	120
3	3	180
4	4	240
5	5	300
6	6	360
7	7	420
8	8	480

9	9	540
10	10	600
11	11	660
12	12	720
13	13	780
14	14	840
15	15	900
16	16	960

- The above table is provided as an example. There are other acceptable concentrations. If there is a risk of volume overload, this dilute solution should only be used as a temporizing measure until a more concentrated solution (eg, 4 micrograms/mL) is available.
- Intravenous epinephrine, like all vasopressors, can cause life-threatening hypertension, cardiac ischemia, and ventricular arrhythmias. It should be administered **only** by clinicians trained and experienced in dose titration of intravenous epinephrine using continuous noninvasive electronic monitoring of heart rate and blood pressure.
- Epinephrine is an ischemia-causing agent and vesicant. Monitor infusion site for extravasation. Central venous catheter administration is preferred when available. Refer to the drug reference included within UpToDate for information on managing extravasation including infiltration of phentolamine.
- To reduce the risk of making a medication error, we suggest that centers have available an institutionally approved protocol for epinephrine infusion that includes steps on how to prepare and administer the infusion by programmable infusion pump and standard concentration(s).

* Unused diluted solutions should be discarded within 24 hours or less of preparation, depending on local standards.

References:

1. *Gahart BL, Nazareno AR, Ortega MQ. Gahart's 2016 Intravenous Medications: A Handbook for Nurses and Health Professionals, 32nd ed, Elsevier-Mosby, St. Louis, MO.*
2. *Lieberman P, Nicklas RA, Oppenheimer J, et al. The diagnosis and management of anaphylaxis practice parameter: 2010 update. J Allergy Clin Immunol 2010; 126:477.*

Graphic 106917 Version 8.0

Table 9: **Example of epinephrine infusion (4 micrograms/mL) - Adult**

Example of preparation of epinephrine infusion for refractory symptoms of anaphylaxis (adult patient) for emergency/critical care units		
Final concentration: Epinephrine 4 micrograms/mL		
Preparation		
1. **CHECK** vial strength.		
2. Add 1 milligram of epinephrine to 250 mL bag of 0.9% normal saline (NS) or 5% dextrose water (D5W) ▪ If you have 0.1 mg/mL strength epinephrine (may also be labeled 1:10,000) use 10 mL to deliver 1 mg to bag* ▪ If you have 1 mg/mL strength epinephrine (may also be labeled 1:1000) use 1 mL to deliver 1 mg to bag*		
Administration		
▪ Start the epinephrine infusion at **0.1 microgram/kg/minute** using a programmable infusion pump while continuously monitoring the patient's cardiac rhythm and blood pressure (ie, approximately 6 to 10 micrograms/minute in most adults). ▪ Using a body weight of 80 kg as an example, you can determine the infusion rate to deliver a dose of 0.1 microgram/kg/minute from a 4 micrograms/mL bag as follows: 1. Multiply 0.1 microgram/kg/minute dose by 80 kg body weight to determine dose in micrograms/minute (= 8 micrograms/minute). 2. Use this table to convert 8 micrograms/minute to infusion rate of 2 mL/minute (= 120 mL/hour). ▪ Every 2 to 3 minutes, increase the infusion rate by approximately one-half of the starting rate until the blood pressure and perfusion improve. For more detail, refer to the UpToDate topic on emergency treatment of anaphylaxis. ▪ In morbidly obese adults, we suggest initiating the infusion using a standard, ie, non-weight-based dose (eg, up to 10 micrograms/minute initially) and titrating to effect.		
Adult infusion dose	**Administration rate for infusion pump to deliver adult dose shown**	
micrograms per minute	**mL per minute**	**mL per hour**
1	0.25	15
2	0.5	30
3	0.75	45
4	1	60
5	1.25	75
6	1.5	90
7	1.75	105
8	2	120

9	2.25	135
10	2.5	150
11	2.75	165
12	3	180
13	3.25	195
14	3.5	210
15	3.75	225
16	4	240

- The above table is provided as an example. There are other acceptable concentrations.
- Intravenous epinephrine, like all vasopressors, can cause life-threatening hypertension, cardiac ischemia, and ventricular arrhythmias. It should be administered **only** by clinicians trained and experienced in dose titration of intravenous epinephrine using continuous noninvasive electronic monitoring of heart rate and blood pressure.
- Epinephrine is an ischemia-causing agent and vesicant. Monitor infusion site for extravasation. Central venous catheter administration is preferred when available. Refer to the drug reference included within UpToDate for information on managing extravasation including infiltration of phentolamine.
- To reduce the risk of making a medication error, we suggest that centers have available an institutionally approved protocol for epinephrine infusion that includes steps on how to prepare and administer the infusion by programmable infusion pump and standard concentration(s).

* Unused diluted solutions should be discarded within 24 hours or less of preparation, depending on local standards.

References:

1. *Gahart BL, Nazareno AR, Ortega MQ. Gahart's 2016 Intravenous Medications: A Handbook for Nurses and Health Professionals, 32nd ed, Elsevier-Mosby, St. Louis, MO.*
2. *Lieberman P, Nicklas RA, Oppenheimer J, et al. The diagnosis and management of anaphylaxis practice parameter: 2010 update. J Allergy Clin Immunol 2010; 126:477.*

Graphic 75682 Version 15.0

Table 10: **Example of epinephrine infusion - Pediatric 10 kg**

Example of preparation of epinephrine infusion for refractory symptoms of anaphylaxis for pediatric patient of 10 kg body weight for emergency/critical care units

Final concentration: Epinephrine 10 micrograms/mL

Preparation

1. **CHECK** vial strength.

2. Add 1 milligram epinephrine to 100 mL bag of 0.9% normal saline (NS) or 5% dextrose water (D5W)
   - If you have 0.1 mg/mL strength epinephrine (may also be labeled 1:10,000) use 10 mL to deliver 1 mg to bag*
   - If you have 1 mg/mL strength epinephrine (may also be labeled 1:1000) use 1 mL to deliver 1 mg to bag*

Administration

- Infuse an initial dose of 0.1 microgram/kg/minute using a programmable infusion pump and titrate, as needed, while continuously monitoring the patient's cardiac rhythm and blood pressure.
- For detail on titrating the infusion based upon response, refer to the UpToDate topic on emergency treatment for anaphylaxis.

Pediatric dose for 10 kg child			**Administration rate for infusion pump to deliver pediatric dose shown**		
microgram per kg per minute		**micrograms per minute**	**mL per *minute* for 10 kg child**		**mL per *hour* for 10 kg child**
0.05	**Multiply by patient weight (10 kg)**	0.5	0.05	**Multiply by 60 minutes**	3
0.1		1	0.1		6
0.2		2	0.2		12
0.3		3	0.3		18
0.4		4	0.4		24
0.5		5	0.5		30
0.6		6	0.6		36
0.7		7	0.7		42
0.8		8	0.8		48
0.9		9	0.9		54
1		10	1		60

- The above table is provided as an example. There are other acceptable concentrations.

- Intravenous epinephrine, like all vasopressors, can cause life-threatening hypertension, cardiac ischemia, and ventricular arrhythmias. It should be administered **only** by clinicians trained and experienced in dose titration of intravenous epinephrine using continuous noninvasive electronic monitoring of heart rate and blood pressure.
- Epinephrine is an ischemia-causing agent and vesicant. Monitor infusion site for extravasation. Central venous catheter administration is preferred when available. Refer to the drug reference included within UpToDate for information on managing extravasation including infiltration of phentolamine.
- To reduce the risk of making a medication error, we suggest that centers have available an institutionally approved protocol for epinephrine infusion that includes steps on how to prepare and administer the infusion and standard concentration(s).

* Unused diluted solutions should be discarded within 24 hours or less of preparation depending on local standards.

References:

1. *Gahart BL, Nazareno AR, Ortega MQ. Gahart's 2016 Intravenous Medications: A Handbook for Nurses and Health Professionals, 32nd ed, Elsevier-Mosby, St. Louis, MO 2016.*
2. *Lieberman P, Nicklas RA, Oppenheimer J, et al. The diagnosis and management of anaphylaxis practice parameter: 2010 update. J Allergy Clin Immunol 2010; 126:477.*

Graphic 56049 Version 14.0

Table 11: **Example of epinephrine infusion - Pediatric 20 kg**

Example of preparation of epinephrine infusion for refractory symptoms of anaphylaxis for pediatric patient of 20 kg body weight for emergency/critical care units

Final concentration: Epinephrine 10 micrograms/mL

Preparation

1. **CHECK** vial strength.

2. Add 1 milligram epinephrine to 100 mL bag of 0.9% normal saline (NS) or 5% dextrose water (D5W)
   - If you have 0.1 mg/mL strength epinephrine (may also be labeled 1:10,000) use 10 mL to deliver 1 mg to bag*
   - If you have 1 mg/mL strength epinephrine (may also be labeled 1:1000) use 1 mL to deliver 1 mg to bag*

Administration

- Infuse an initial dose of 0.1 microgram/kg/minute using a programmable infusion pump and titrate, as needed, while continuously monitoring the patient's cardiac rhythm and blood pressure.
- For detail on titrating the infusion based upon response, refer to the UpToDate topic on emergency treatment of anaphylaxis.

Pediatric dose for 20 kg child			**Administration rate for infusion pump to deliver pediatric dose shown**		
microgram per kg per minute		**micrograms per minute**	**mL per *minute* for 20 kg child**		**mL per *hour* for 20 kg child**
0.05	**Multiply by patient weight (20 kg)**	1	0.1	**Multiply by 60 minutes**	6
0.1		2	0.2		12
0.2		4	0.4		24
0.3		6	0.6		36
0.4		8	0.8		48
0.5		10	1		60
0.6		12	1.2		72
0.7		14	1.4		84
0.8		16	1.6		96
0.9		18	1.8		108
1		20	2		120

- The above table is provided as an example. There are other acceptable concentrations.

- Intravenous epinephrine, like all vasopressors, can cause life-threatening hypertension, cardiac ischemia, and ventricular arrhythmias. It should be administered **only** by clinicians trained and experienced in dose titration of intravenous epinephrine using continuous noninvasive electronic monitoring of heart rate and blood pressure.
- Epinephrine is an ischemia-causing agent and vesicant. Monitor infusion site for extravasation. Central venous catheter administration is preferred when available. Refer to the drug reference included within UpToDate for information on managing extravasation including infiltration of phentolamine.
- To reduce the risk of making a medication error, we suggest that centers have available an institutionally approved protocol for epinephrine infusion that includes steps on how to prepare and administer the infusion and standard concentration(s).

* Unused diluted solutions should be discarded within 24 hours or less of preparation, depending on local standards.

References:

1. *Gahart BL, Nazareno AR, Ortega MQ. Gahart's 2016 Intravenous Medications: A Handbook for Nurses and Health Professionals, 32nd ed, Elsevier-Mosby, St. Louis, MO 2016.*
2. *Lieberman P, Nicklas RA, Oppenheimer J, et al. The diagnosis and management of anaphylaxis practice parameter: 2010 update. J Allergy Clin Immunol 2010; 126:477.*

Graphic 66791 Version 14.0

Table 12: **Beneficial effects and adverse effects of epinephrine in the treatment of anaphylaxis**

Beneficial effects	
At alpha-1 receptor:	
Increased vasoconstriction (at low doses)	
Increased peripheral vascular resistance	
Increased blood pressure	
Decreased mucosal edema (eg, in larynx)	
At beta-1 receptor:	
Increased heart rate (chronotropy)	
Increased force of cardiac contraction (inotropy)	
At beta-2 receptor:	
Decreased mediator release from mast cells and basophils	
Increased bronchodilation	
Increased vasodilation	
Adverse effects*¶Δ	
Common and transient	Anxiety, palpitations, pallor, tremor, fear, restlessness, dizziness, headache
Uncommon (typically occur after overdose)	Ventricular arrhythmias, angina, myocardial infarction, pulmonary edema, sudden sharp increase in blood pressure, intracranial hemorrhage

* Risk of adverse effects may be increased in the following conditions:
- Use of tricyclic antidepressants, monoamine oxidase inhibitors, or cocaine.
- Some preexisting cardiovascular, central nervous system, or thyroid diseases. Examples include intracranial surgery, acute aneurysm, or untreated hyperthyroidism.

¶ Serious adverse effects such as those listed in the table potentially occur when epinephrine is given in overdose by any route, most commonly after an intravenous bolus injection, an overly rapid intravenous infusion, or an erroneous intravenous injection of a 1 mg/mL (1:1000) epinephrine solution instead of an appropriately diluted 0.1 mg/mL (1:10,000) or a 0.01 mg/mL epinephrine solution.

Δ Anaphylaxis can present as an acute coronary syndrome, arrhythmias, myocardial infarction, or angina, before or in the absence of epinephrine injection. This potentially occurs in patients with known coronary artery disease, patients in whom subclinical coronary artery disease is unmasked, and patients (including children) with transient coronary artery vasospasm in whom no cardiovascular abnormalities can be detected by electrocardiogram or echocardiography after recovery from anaphylaxis.

Graphic 65684 Version 15.0

Table 13: **Dosing for commonly used, minimally sedating oral antihistamines**

Medication	Adult dosing	Pediatric dosing
Cetirizine*¶	**Standard:** 10 mg once daily **New-onset urticaria:** May increase to 10 mg twice daily **Chronic urticaria:** May gradually increase to 20 mg twice daily	**≥12 years:** 10 mg once daily **≥6 to 11 years:** 5 to 10 mg once daily **2 to <6 years:** 2.5 to 5 mg once daily or 2.5 mg twice daily **6 months to 2 years:** 2.5 mg once daily NOTE: For chronic urticaria and perennial rhinitis, dose may be increased to 2.5 mg twice daily for children aged 12 months to 2 years
Levocetirizine*	**Standard:** 5 mg once daily **New-onset urticaria:** May increase to 5 mg twice daily **Chronic urticaria:** May gradually increase to 10 mg twice daily	**≥12 years:** 2.5 to 5 mg once daily **6 to 11 years:** 2.5 mg once daily **6 months to <6 years:** 1.25 mg once daily
Loratadine¶	**Standard:** 10 mg once daily **New-onset urticaria:** May increase to 10 mg twice daily **Chronic urticaria:** May gradually increase to 20 mg twice daily	**≥6 years:** 10 mg once daily or in 2 divided doses **2 to <6 years:** 5 mg once daily
Desloratadine¶	**Standard:** 5 mg once daily **New-onset urticaria:** May increase to 5 mg twice daily **Chronic urticaria:** May gradually increase to 10 mg twice daily	**≥12 years:** 5 mg once daily **6 to 11 years:** 2.5 mg once daily **1 to <6 years:** 1.25 mg once daily (requires oral solution; not commercially available in US) **6 months to 1 year:** 1 mg once daily (requires oral solution; not commercially available in US)
Fexofenadine¶	**Standard:** ▪ 24-hour formulation: 180 mg once daily ▪ 12-hour formulation: 60 mg twice daily **New-onset or chronic urticaria:** ▪ 24-hour formulation: 180 mg once daily (higher doses are typically not more effective)	**≥12 years:** ▪ 24-hour formulation: 180 mg once daily ▪ 12-hour formulation: 60 mg twice daily **2 to 11 years:** 30 mg twice daily **6 months to 2 years and ≥10.5 kg:** 30 mg twice daily **6 months to 2 years and <10.5 kg:** 15 mg twice daily

Ebastine[Δ]	**Standard:** 10 mg once daily **New-onset or chronic urticaria:** May increase to 20 mg once daily	**≥12 years:** 10 mg once daily
Bilastine[Δ]	**Standard:** 20 mg once daily **New-onset or chronic urticaria:** May gradually increase to 80 mg once daily	**≥12 years:** 20 mg once daily **≥4 to 11 years and ≥16 kg:** 10 mg once daily
Rupatadine[Δ]	**Standard:** 10 mg once daily **New-onset or chronic urticaria:** May gradually increase to 40 mg once daily	**≥12 years:** 10 mg once daily **≥2 to 11 years and >25 kg:** 5 mg once daily **≥2 to 11 years and 10 to ≤25 kg:** 2.5 mg once daily

This table is to be used with UpToDate content on the treatment of allergic rhinitis, new-onset urticaria (hives), chronic spontaneous urticaria, and other allergic disorders. Standard dosing is used for treatment of allergic rhinitis and as initial dosing for urticaria. For chronic conditions, regular (daily) use is generally recommended; however, as-needed use may be sufficient for mild symptoms. Doses listed are for patients with normal kidney and liver function. Doses may need to be modified in kidney or hepatic impairment; refer to UpToDate Lexidrug monographs for details. Significant drug interactions may exist requiring dose adjustment or avoidance of some combinations; refer to the drug interactions program included within UpToDate to determine specific interactions.

IV: intravenous.

* Cetirizine and levocetirizine may be more sedating than others listed; they can be taken in the evening if sedating. Cetirizine is also available as an IV solution that may be useful in a health care setting when a rapid onset is needed or oral administration is not available.

¶ Also available as a single-pill combination with pseudoephedrine.

Δ Widely available in areas other than United States.

Graphic 148024 Version 5.0

Table 14: **Types of ECLS in the intensive care unit**

ECLS type	ECLS mode and configuration	Organ support and function	Suggested indications*
ECMO	VV ECMO	Respiratory support (oxygenation and CO_2 removal)	Refractory ARDS, bridge to lung transplant, severe primary graft dysfunction after lung transplant, severe reperfusion pulmonary edema after thromboendarterectomy, aspiration
	VA ECMO	Cardiac/circulatory support (circulatory support, oxygenation and CO_2 removal)	Refractory cardiogenic shock, ECPR, decompensated pulmonary vascular disease, postcardiotomy shock, others (trauma, anaphylactic shock, drowning, organ donation, shock due to overdose, hypothermia)
	VVA ECMO VAV ECMO	Cardiorespiratory support	Mixed components of the above indications
$ECCO_2R$	VV $ECCO_2R$ AV $ECCO_2R$	CO_2 removal only	Refractory hypercapnic respiratory failure (eg, status asthmaticus, bridge to lung transplantation); facilitation of lung protective ventilation strategies in ARDS (under investigation)

NOTE: This table excludes cardiopulmonary bypass for cardiothoracic or vascular surgery.

ARDS: acute respiratory distress syndrome; AV: arteriovenous; CO_2: carbon dioxide; $ECCO_2R$: extracorporeal carbon dioxide removal; ECLS: extracorporeal life support; ECMO: extracorporeal membrane oxygenation; ECPR: extracorporeal cardiopulmonary resuscitation; VA: venoarterial; VAV: venoarteriovenous; VV: venovenous; VVA: venovenoarterial.

* In most instances, ECMO is offered when standard therapies have failed.

Adapted from: Conrad SA, Broman LM, Taccone FS, et al. The Extracorporeal Life Support Organization Maastricht Treaty for nomenclature in extracorporeal life support. A position paper of the Extracorporeal Life Support Organization. Am J Respir Crit Care Med 2018; 198:447.

Graphic 140283 Version 2.0

Table 15: **Possible risk factors for biphasic anaphylaxis**

More certain (consistently associated in studies)
▪ Severe initial presentation (eg, hypotension or severe respiratory compromise) ▪ Delayed time to first epinephrine treatment (eg, >60 minutes) ▪ Need for >1 dose of epinephrine
Less certain (implicated in some studies but not others)
▪ Unknown trigger ▪ Possible ongoing exposure to trigger: sustained-release medications or depot medications, foods ▪ A prolonged interval (eg, hours) between contact with the trigger and the start of initial symptoms ▪ Treatment with antihistamines or glucocorticoids without epinephrine

The table lists possible risk factors for biphasic anaphylaxis based on several meta-analyses and studies. Refer to UpToDate topics on anaphylaxis for recommendations on duration of observation in patients with anaphylaxis and one or more risk factors.

References:

1. *Shaker MS, Wallace DV, Golden DBK, et al. Anaphylaxis—a 2020 practice parameter update, systematic review, and Grading of Recommendations, Assessment, Development and Evaluation (GRADE) analysis. J Allergy Clin Immunol 2020; 145:1082.*
2. *Lee S, Bellolio MF, Hess EP, et al. Time of onset and predictors of biphasic anaphylactic reactions: A systematic review and meta-analysis. J Allergy Clin Immunol Pract 2015; 3:408.*
3. *Kraft M, Hofmeier KS, Ruëff F, et al. Risk factors and characteristics of biphasic anaphylaxis. J Allergy Clin Immunol Pract 2020; 8:3388.*
4. *Liu X, Lee S, Lohse CM, et al. Biphasic reactions in emergency department anaphylaxis patients: A prospective cohort study. J Allergy Clin Immunol Pract 2020; 8:1230.*

Graphic 148698 Version 1.0

Contributor Disclosures

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